Force

Force : Push or Pull , Newton (N) ≈ the force it takes to hold up a small apple (~100g) against the gravity

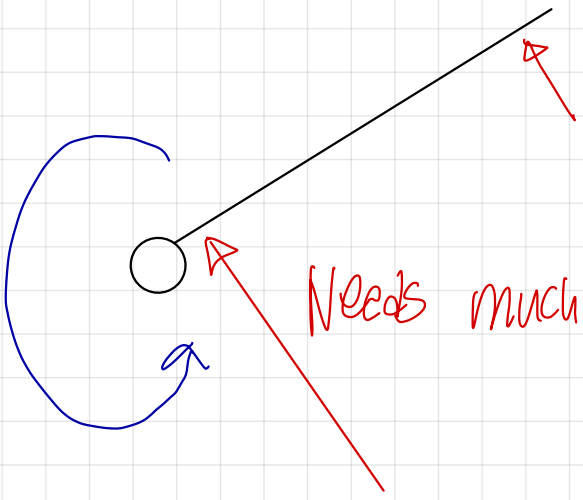
(kg · m/s²) (kg) (m/s²)

Force = mass × acceleration

$F = ma$

Torque (2D)

Torque : a Twisting force , the rotational version of regular push/pull force



(N · m) (N) (m)

torque = force × distance from pivot

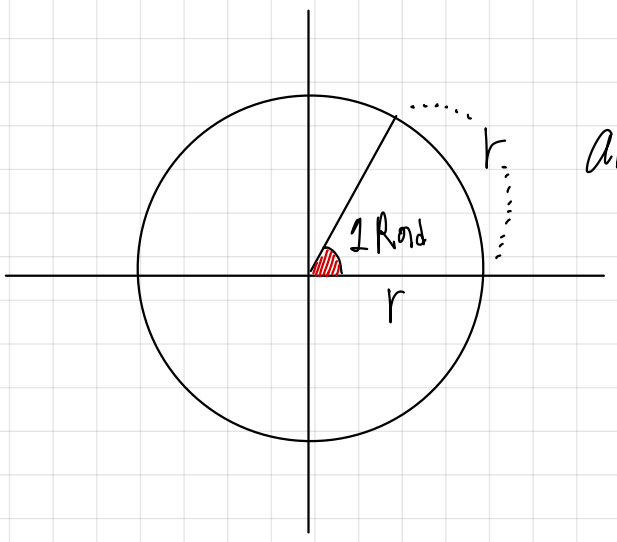
= lever arm

= moment arm

$\tau = Fr$

When the push is perpendicular to the lever

Radian



s
arc length

(m) (m) (unitless)

arc length = radius × angle in radians

$s = r\theta$

$\theta = \frac{s}{r}$

- A full circle : $360^\circ = 2\pi$ radians

$$\theta(\text{rad}) = \theta(\text{deg}) \times \frac{2\pi}{360}$$

$$\theta(\text{deg}) = \theta(\text{rad}) \times \frac{360}{2\pi}$$

- π = Circumference / diameter (unitless) (m) (m)

of ANY Circle

$\pi = C/d \approx 3.14$

- $C = \pi \times d = \pi \times 2r = 2\pi r$

$= (2\pi)\, r$

If arc length is C (the whole circle) then

$S = \theta\, r \Rightarrow \theta = (2\pi)$

Angular Velocity and Acceleration

(m/s)

$$\text{Velocity} = \frac{\Delta \text{ position (m)}}{\Delta \text{ time (s)}}$$

$$V = \frac{\Delta x}{\Delta t}$$

How much distance per unit time

(rad/s)

$$\text{Angular velocity} = \frac{\Delta \text{ angle (in rad)}}{\Delta \text{ time (s)}}$$

$$\omega = \frac{\Delta \theta}{\Delta t}$$

How much angle per unit time

(m/s²)

$$\text{Acceleration} = \frac{\Delta \text{ velocity (m/s)}}{\Delta \text{ time (s)}}$$

$$a = \frac{\Delta V}{\Delta t}$$

How fast velocity changes per unit time

(rad/s²)

$$\text{Angular acceleration} = \frac{\Delta \text{ angular velocity (rad/s)}}{\Delta \text{ time (s)}}$$

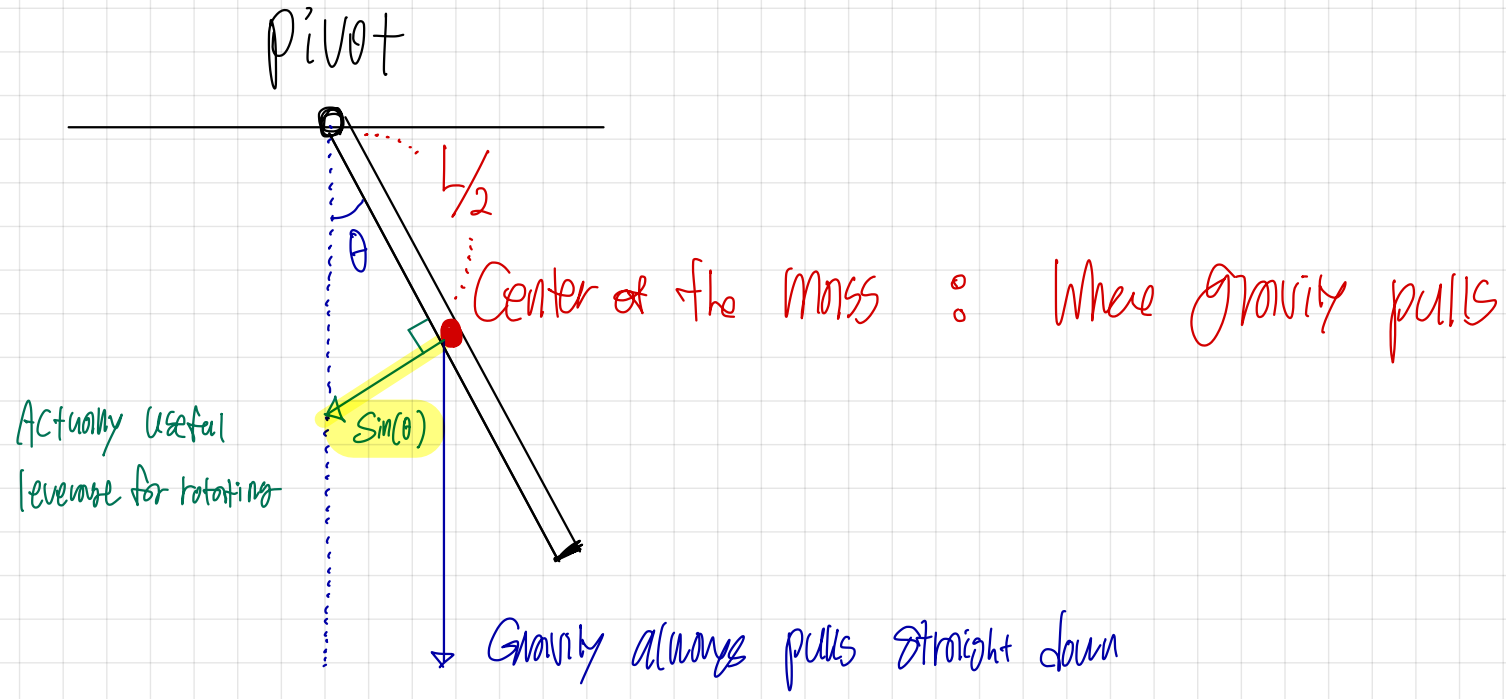
$$\alpha = \frac{\Delta \omega}{\Delta t}$$

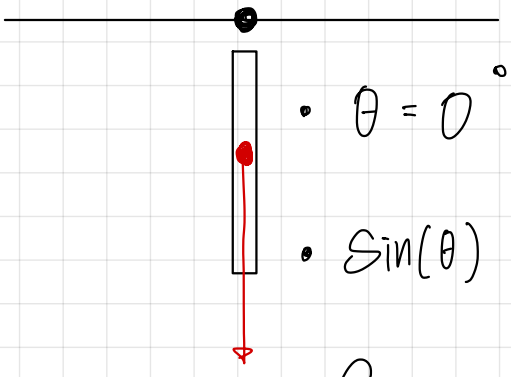
How fast angular velocity changes per unit time

Gravitational Torque

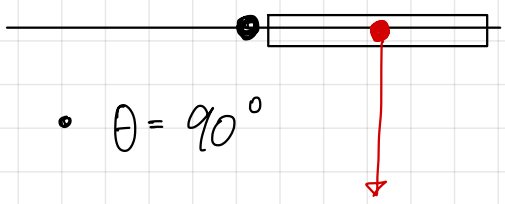
Gravitational Torque : the torque that gravity creates around the pivot trying to swing the pendulum

↳ torque $\tau = F \times r$ (moment arm)





- $\theta = 0°$
- $\sin(\theta) = 0$ (No use)
- Gravity is doing No work for rotation



- $\theta = 90°$
- $\sin(\theta) = 1$ (full use)
- Gravity is doing full work for rotation

Moment Arm : the perpendicular distance from the pivot to the line along with the force acts

(m) (m) (dimensionless)

$$\text{Moment arm} = L/2 \times \sin(\theta)$$

(N·m) (N) (m)

$$\tau_{gravity} = F_{gravity} \times \text{Moment Arm}$$

$$= (m \times g) \times (L/2 \times \sin(\theta))$$

(kg) (m/s^2) (m) (unitless)

Damping Torque

Damping : Any force or torque that opposes the direction of current motion, grows in strength as the speed of motion increases, and is zero when there is no motion at all

1) Viscous damping (linear)

(N·m) ($N \cdot m \cdot s/rad$) (rad/s)

$$T = -C \times \omega$$

Smooth, proportional to speed, realistic for air resistance or fluid-like resistance in a joint

2) Coulomb Damping (Dry friction or Stiction)

(N·m) (N·m) (unitless)

$$T = -\mu \times Sign(\omega)$$

A Constant resistance, same strength no matter how fast

3) Quadratic Damping (air resistance at high speed)

(N·m) ($N \cdot m \cdot s^2/rad^2$) (rad/s) (rad/s)

$$T = -C \times \omega \times |\omega|$$

↳ I_{total} for the pendulum model)

$$I_{total} = I_{gravity} + I_{viscous} + I_{coulomb}$$

Inertia 2D about one point at one distance

Inertia : an Object's natural resistance to having its motion Changed (Newton's first law)

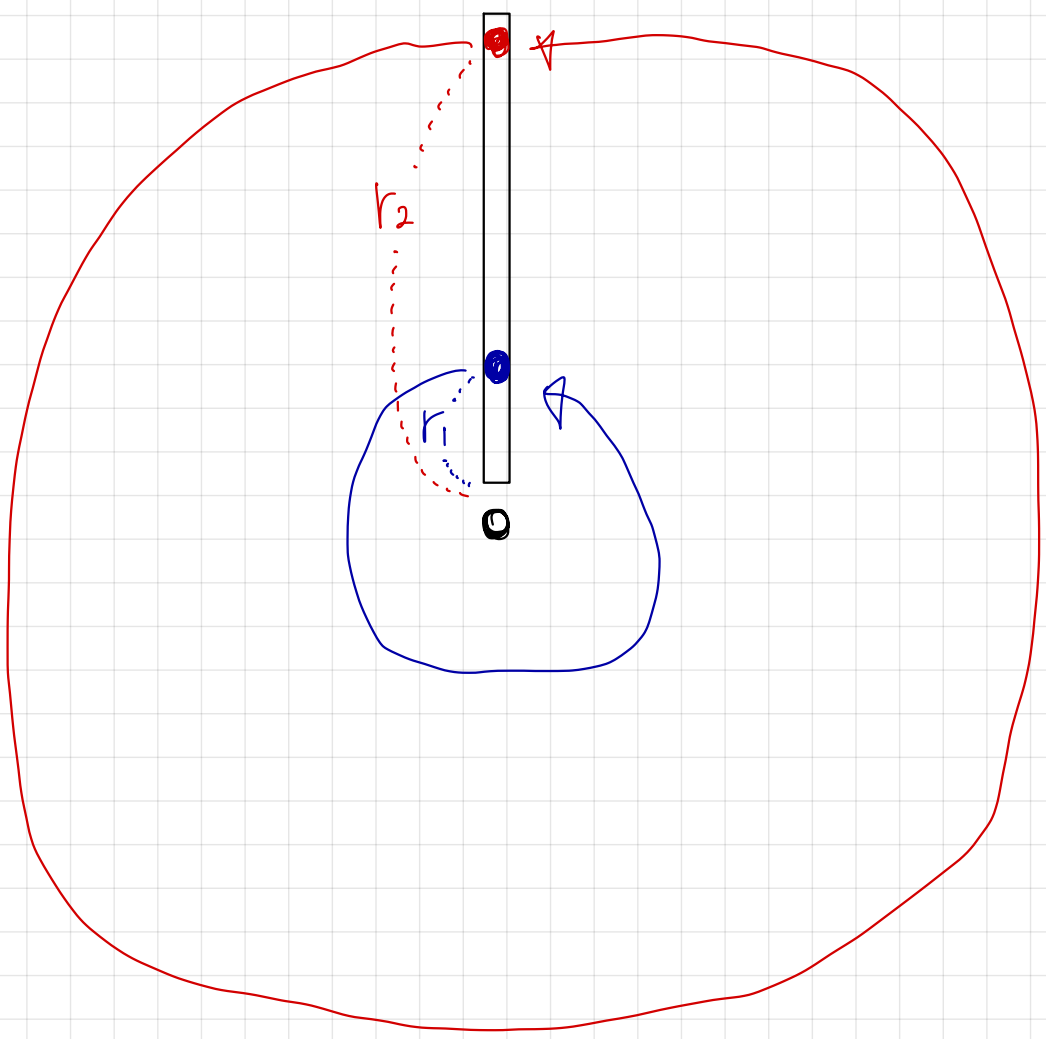
↳ Straight line motion : Inertia = Mass (kg)

$$F = m\,a$$

Linear Motion (translational motion) Not for rotational Motion

↳ Why Chunks ?

Why rotational Motion Needs Something More than Mass?



Both points has the same amounts of time to rotate

A point sitting far away from the pivot should travel much further thus require higher speed

⇒ more expensive to spin

$(kg \cdot m^2)$ (kg) (m)

$$I = m \cdot r^2$$

(One point mass at one distance

Not w.r.t a full rod)

0) If we push a mass on the end of a rod to make it spin faster, how much push is needed in terms of angular quantities?

1) $S = r \times \theta \Rightarrow \frac{dS}{dt} = r \times \frac{d\theta}{dt}$

Constant $\frac{d\theta}{dt}$: ω : angular velocity

↳ the distance if the mass sweeps through some angle θ

2) Velocity is defined as how fast distance changes over time

$$V = \frac{dS}{dt} = r\omega \quad , \quad \frac{dv}{dt} = r \cdot \frac{d\omega}{dt}$$

Constant $\frac{d\omega}{dt}$: α : angular acceleration

3) Acceleration : how fast velocity changes over time

$$a = \frac{dv}{dt} = r\alpha$$

4) Newton's second Law

$$F = m \times a = m \times r\alpha$$

$$= \underbrace{m\alpha}_{F} \times r$$

5) Torque

$$\tau = F \times r = m \times r \times \alpha \times r = \underbrace{mr^2}_{I}\alpha$$

6) General Rule for Rotation (N2L for rotation)

Inertia angular acceleration

$$\therefore \quad \tau_{net} = I \times \alpha$$

$(N \cdot m)$ $(kg \cdot m^2)$ (rad/s^2)

$$\tau_1 = I_1 = m_1 \times r_1^2 \times \alpha$$

$$\tau_2 = I_2 = m_2 \times r_2^2 \times \alpha$$

$$\vdots$$

$$\tau_n = I_n = m_n \times r_n^2 \times \alpha$$

$$\tau_{total} = \tau_1 + \tau_2 + \dots + \tau_n = \alpha \times \underbrace{\sum m_i r_i^2}_{I}$$

$(kg \cdot m^2)$ $(kg \cdot m^2)$

$$\therefore \quad I = \frac{\tau}{\alpha} = mr^2$$

↳ Discrete Chunks) $I = \sum m_i r_i^2$

Continuous Chunk) $I = \int_0^L r^2 \, dm$

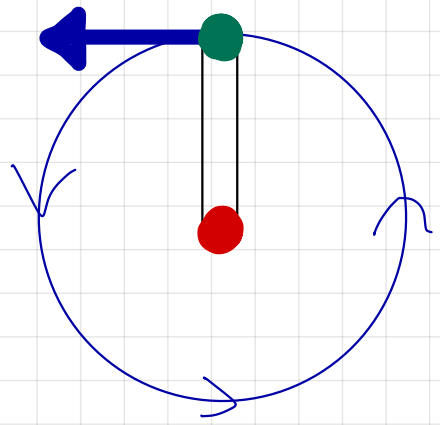
- Interpretation) $I = T / \alpha$

I is the amount of torque required to produce One unit of angular acceleration

⇒ If $I = 0.083\ kg \cdot m^2$ for the pendulum, to make this pendulum accelerate angularly at 1 rad/s^2, we need exactly 0.083 $N \cdot m$ of torque

Inertia about the pivot (end) across the rod

$$I = \frac{M \cdot L^2}{3} = \text{Rod mass} \cdot \text{Rod Length}^2 / 3$$



- Derivation

i) mass per unit length

Total mass of the rod : M $[kg]$

Total length of the rod : L $[m]$

mass per unit length : M/L $[kg/m]$

ii) tiny slice

dr : a very small length

$\Rightarrow$ How much mass in that sliver?

dm = mass per unit length $\times$ dr

$dm = (M/L) \times dr \Rightarrow$ one chunk mass

(ii) Contribution

Contribution depends on the moment arm r

$$\text{Contribution} = dm \times r^2$$

$$= \left(\frac{m}{L}\right) dr \times r^2$$

↳ one slice's contribution

iv) Total Contribution (since the origin to the end : 0 to L)

$$I = \int_0^L r^2 \left(\frac{m}{L}\, dr\right)$$

$$= \frac{m}{L} \int_0^L r^2 dr$$

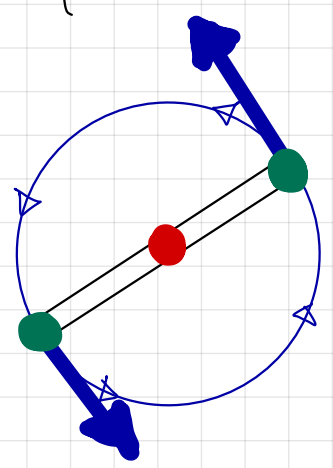
$$= \frac{m}{L} \left(\frac{r^3}{3}\Big|_{r=L} - \frac{r^3}{3}\Big|_{r=0} \right)$$

$$= \frac{m}{L} \cdot \frac{L^3}{3}$$

$$I = \frac{1}{3} m L^2$$

Inertia about the Center of Mass across the rod

How hard is it to change the rotational speed of the rod (when it spins around an axis) through the center of mass, perpendicular



$$I_{COM} = \frac{m \cdot L^2}{12} = \text{rod mass} \cdot \text{rod length}^2 \,/\, 12$$

$(kg \cdot m^2)$ (kg) (m)

- How hard to change the rotational speed (det of Inertia):

↳ means to speed up or down the spinning = angular acceleration $\left(\alpha = rad/s^2\right)$

↳ how hard = how much torque $\left(\tau = I \cdot \alpha = (N \cdot s) = (kg \cdot m/s) \cdot (rad/s^2)\right)$

so Inertia is torque needed per unit of angular acceleration $\left(I = \frac{\tau}{\alpha}\right)$

ex) If $I_{COM} = 0.0833\ kg \cdot m^2$ then to speed the rod's spin up by 1 rad/s^2 each second

we need 0.0833 N·s of torque

- When it spins about an axis:

↳ Axis of Rotation: the straight line that stays still while everything else turns around it

↳ With the torque fixed, a smaller moment of inertia gives a bigger angular acceleration

(m) (m)

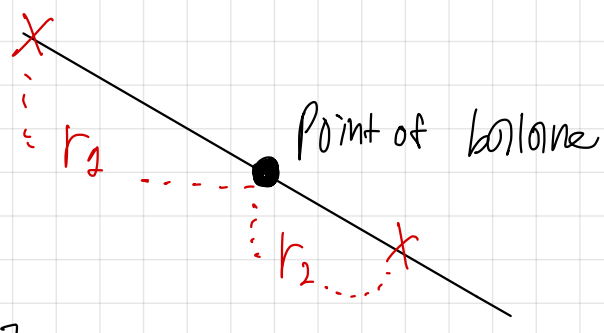
↳ arc length : $S = r \times \theta$

radius angle

• Derivation) a uniform rod, mass m (kg), length L (m),
axis through the center of mass perpendicular to the rod

1) inertia = $\int r^2 dm$ (r : the distance of each piece from the axis)

2) mass per unit length = m/L [kg/m]



mass of a slice : $dm = (m/L)\, dr$ [kg]

limits : r from $-L/2$ to $L/2$ [m] $\Rightarrow \int_{-L/2}^{L/2}$

↳ about the COM : r is measured from the balance point

about the end : r is measured from the end

3) Inertia) $I_{com} = \int_{-L/2}^{L/2} r^2 \cdot \left[(m/L)\, dr \right]$

$$= \frac{m}{L} \int_{-L/2}^{L/2} r^2 \, dr$$

$$= \frac{m}{L} \left[\frac{r^3}{3} \right]_{-L/2}^{L/2}$$

$$= \frac{m}{L}\left(\frac{(L/2)^3}{3} - \frac{(-L/2)^3}{3}\right)$$

$$= \frac{mL^3}{12}$$

Tangential Velocity in 2D

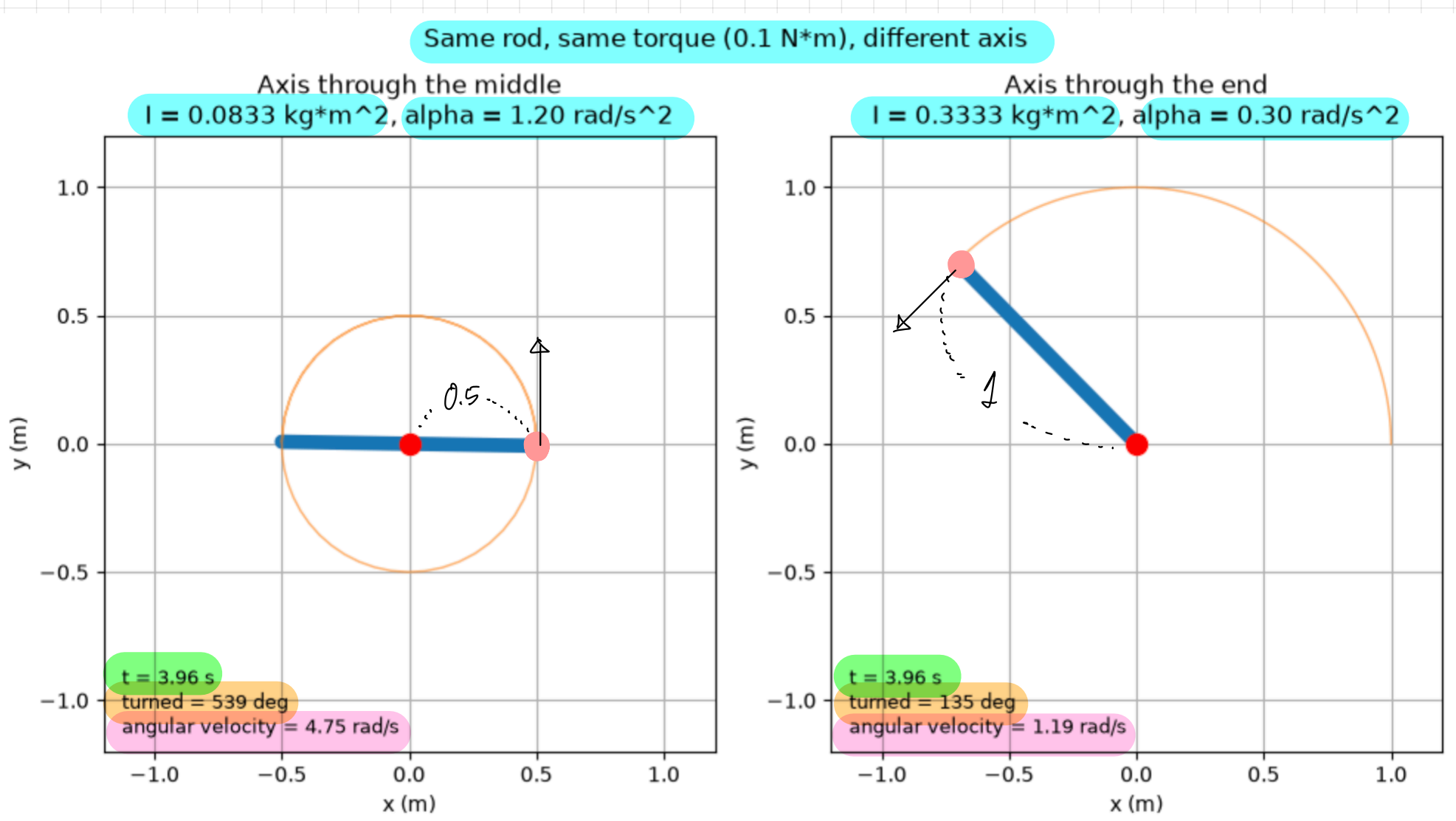
Linear speed/velocity of one point on the rod :

$$\text{Speed} \quad \underbrace{v = r \times \omega}_{\text{radius} \quad \text{angular velocity}} = r \times \frac{\theta}{s}$$

$\left(\frac{m}{s}\right) \quad (m) \quad \left(\frac{rad}{s} = \frac{1}{s}\right)$

$\Rightarrow$ Speed = distance traveled per unit time

$\Rightarrow$ angular velocity ω = Change in theta / Change in time



ω r

$\hookrightarrow$ Speed of the tip = $(0.5) \times (4.75)$

$= 2.4 \text{ m/s}$

ω r

$\hookrightarrow$ Speed of the tip = $(1) \times (1.19)$

$= 1.2 \text{ m/s}$

↳ the direction of V : Tangent to its circle (points along the circle, right angle to the rod)

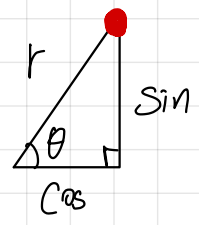
Why ?)

The position of the tip when the rod has turned by angle θ (rad) with θ changing at rate w :

(m)

$$\text{position} = \left(r \cdot \cos(\theta),\ r \cdot \sin(\theta) \right)$$

$\vec{r}$



The velocity which is the rate of change of position. Differentiate w.r.t time.

(m/s)

Since $\frac{d\theta}{dt} = w$:

$$\text{velocity} = \left(-r \cdot w \cdot \sin(\theta),\ r \cdot w \cdot \cos(\theta) \right)$$

$\vec{v}$ ↳ $(r \cdot \cos(\theta))'$ ↳ $(r \cdot \sin(\theta))'$

Check the size :

$$\text{Speed} = \sqrt{(-rw\sin(\theta))^2 + (rw\cos(\theta))^2}$$

$$= r \cdot w \cdot \sqrt{\sin^2\theta + \cos^2\theta}$$

$$= r \cdot w \cdot 1 = r \cdot w$$

Check the direction : Dot Product of Position Arrow and Velocity Arrow

↳ dot product of $a = (a_x, a_y)$ and $b = (b_x, b_y)$

$$a \cdot b = a_x b_x + a_y b_y$$

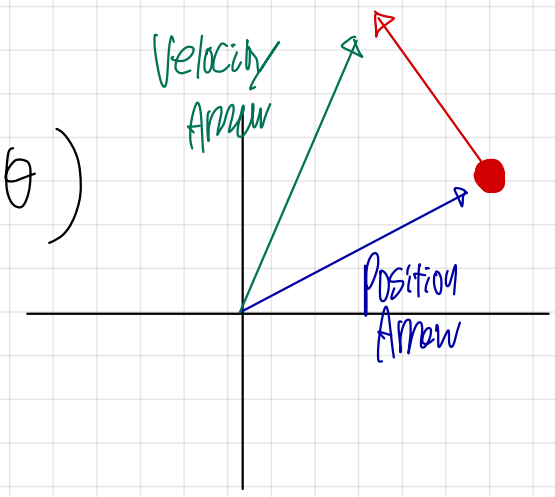
dot

Shows how aligned a and b

$$= (\text{length of } a) \times (\text{length of } b) \times \cos(\theta)$$

$\Rightarrow$ Position Arrow : $(r\cos\theta, r\sin\theta)$

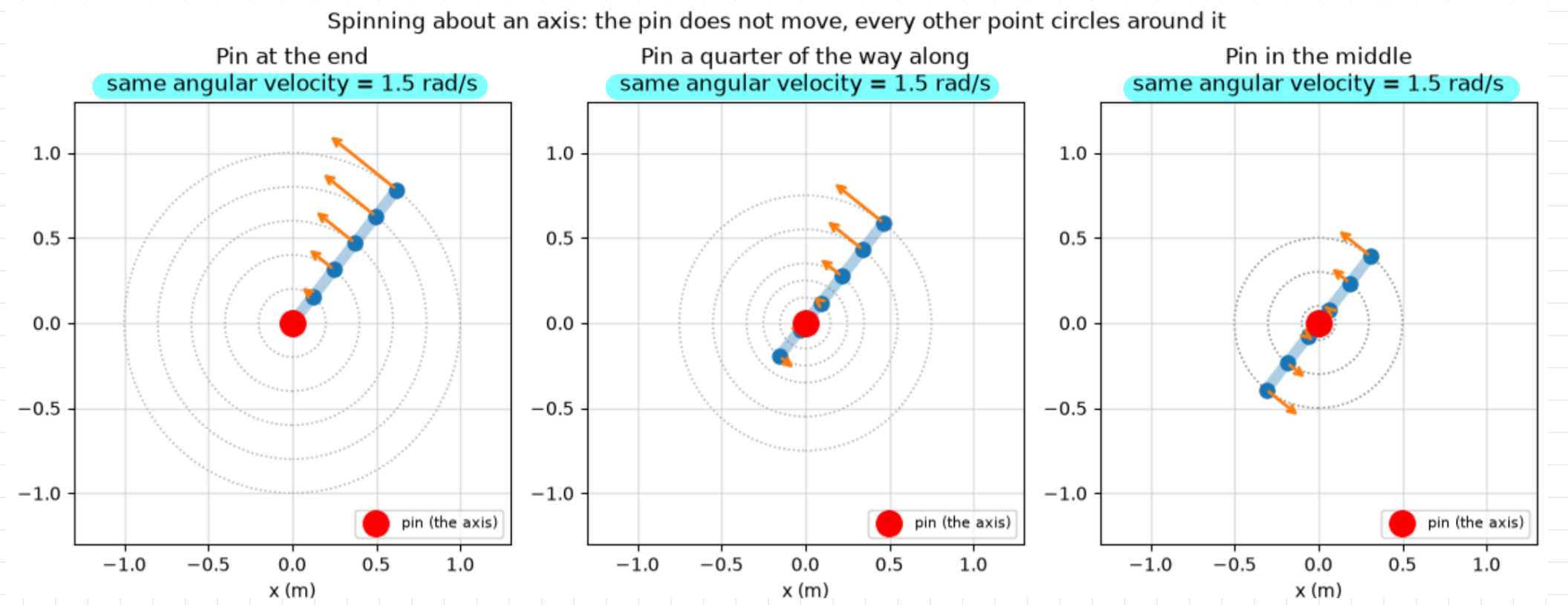
$\Rightarrow$ Velocity Arrow : $(-r\omega\sin\theta, r\omega\cos\theta)$



$$PA \cdot VA = (r\cos\theta \cdot -r\omega\sin\theta) + (r\sin\theta \cdot r\omega\cos\theta)$$

$$= -r^2\omega\sin\theta\cos\theta + r^2\omega\sin\theta\cos\theta$$

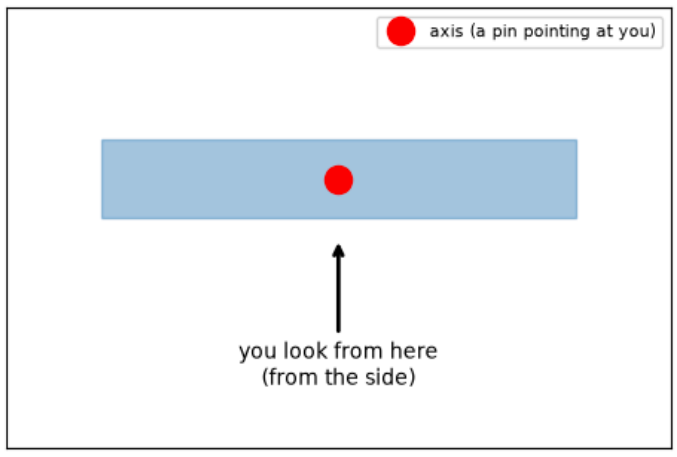
$$= 0$$ thus perpendicular to the position Arrow



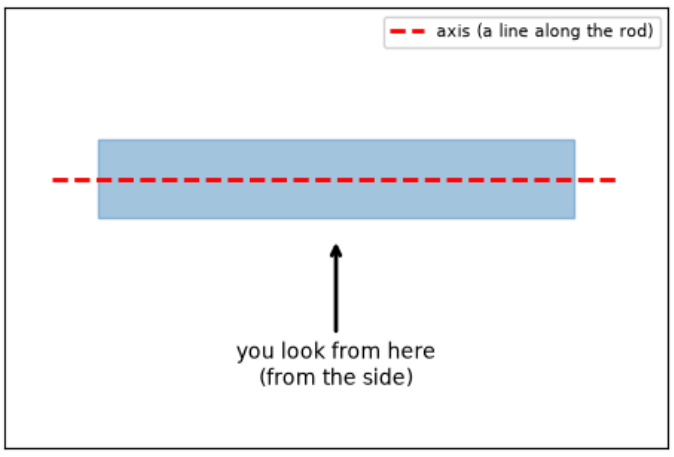
Rod Viewpoints

Same rod, different place to stand (not to scale: the rod is drawn thicker than it is)

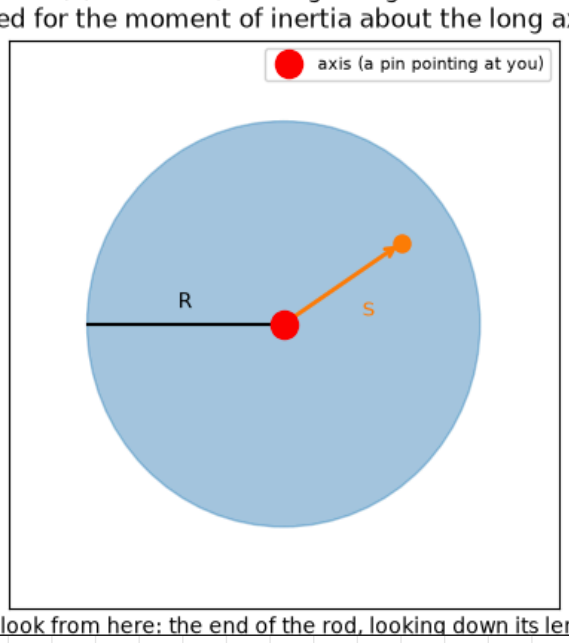
(a) Side view
used for the moment of inertia about the center of mass



(b) The same side view, but the axis runs along the rod
from here the axis is a LINE, not a dot



(c) End view, looking along the axis
used for the moment of inertia about the long axis



you look from here: the end of the rod, looking down its length

Side View

Side View

End View / Axial View
(Viewed along the axis)

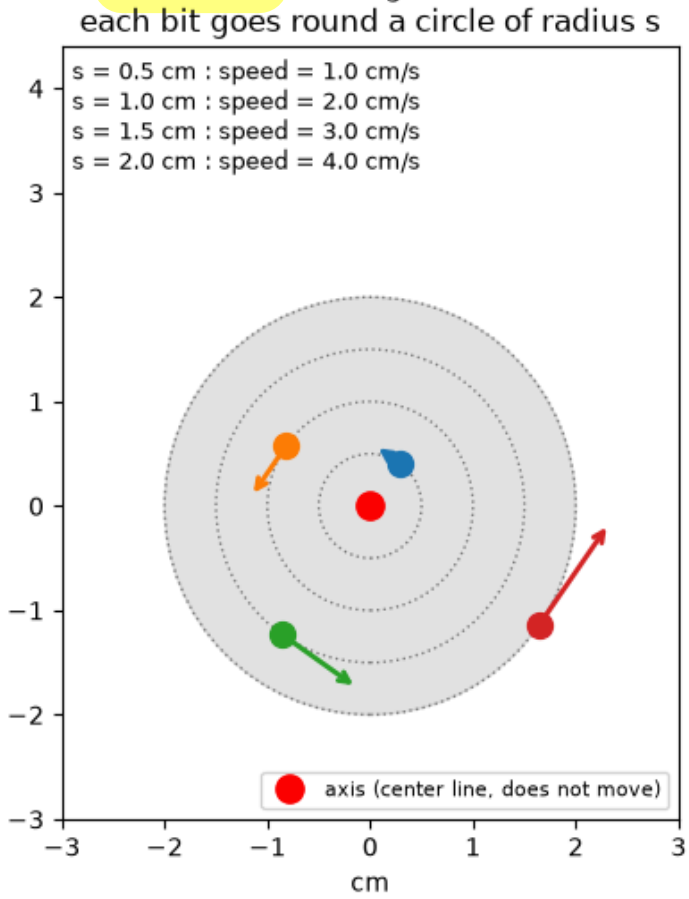
Inertia about the Z axis (I_{zz})

Principal Moment of Inertia (twirl)

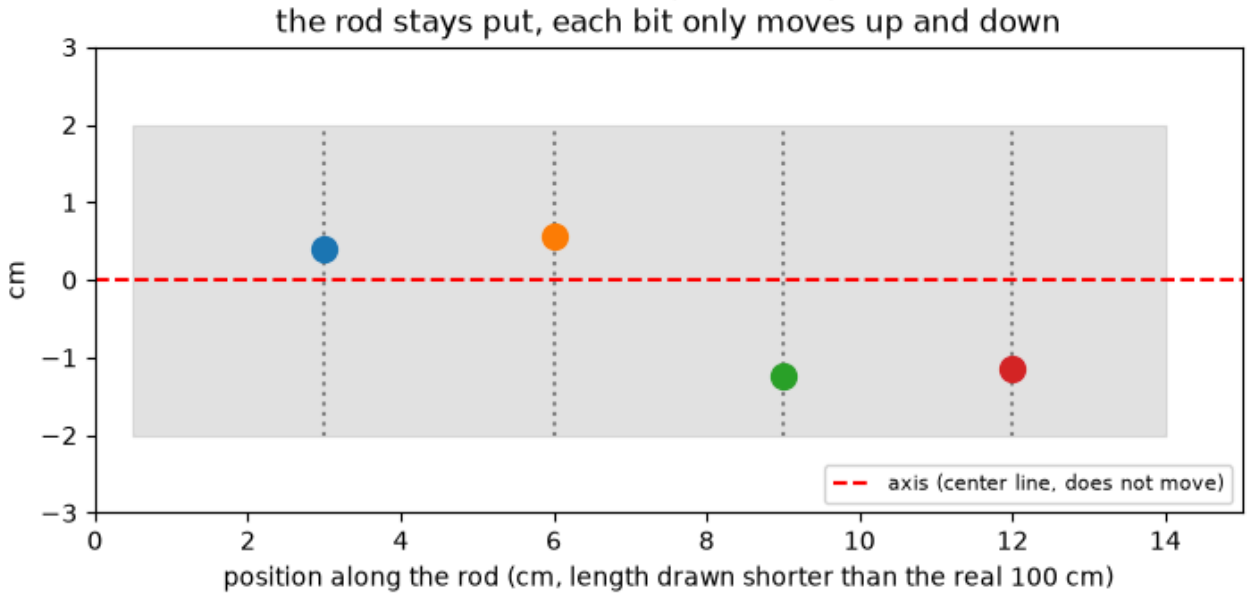
↳ A number that says how stubborn the rod is against being twirled about its own length

Twirling the rod about its own long axis, angular velocity = 2.0 rad/s

END VIEW (looking down the rod)
each bit goes round a circle of radius s



SIDE VIEW (same bits)
the rod stays put, each bit only moves up and down



$(kg \cdot m^2)$ (kg) (m)

$$I_{zz} = \frac{m \cdot R^2}{2} = \text{Rod mass} \cdot \text{Rod Radius}^2 \,/\, 2$$

- Derivation

1) Inertia : $I = \sum_i \Delta m_i r_i^2 = \int r^2 \, dm$

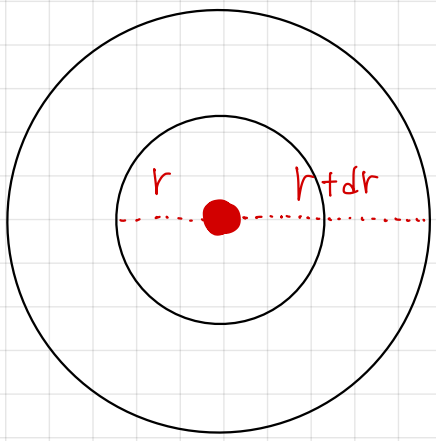
2) Mass per unit volume : R : rod radius , L : rod length , m : rod mass

Volume of the cyclinder $V = \pi R^2 L$ $[m^3]$

Density $\rho = \frac{m}{V} = \frac{m}{\pi R^2 L}$ $[kg/m^3]$

Since rod is uniform , the density is the same everywhere

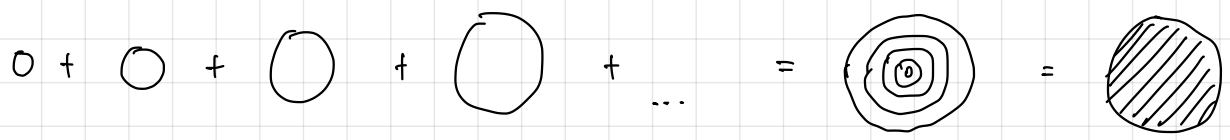
3) Volume of one ring : dV



the rings lies between distance r and r+dr

and it is as long as the rod

So the volume is (area seen from the end) x L



L, we can control the size of R : start from r=0 to r=R

↳ Volume of one ring:

i) $\pi(r+dr)^2 - \pi r^2 = 2\pi r\,dr + \pi\,dr^2$

ii) dr is very small so dr^2 is negligible

$$dV = 2\pi r\,dr \cdot L \quad (m^3)$$

4) Mass of one ring : dm

$$dm = \rho\,dV \quad (\text{since } \rho = \frac{m}{V})$$

$$= \rho \cdot 2\pi r\,dr \cdot L$$

$$= \frac{m}{\pi R^2 L} \cdot 2\pi r\,dr \cdot L$$

$$dm = \frac{2m}{R^2} r\,dr \quad [kg]$$

↳ All rings together should give m : $\int_0^R \frac{2m}{R^2} r\,dr = \frac{2m}{R^2}\frac{R^2}{2} = m$

5) Substitute

$$I_{zz} = \int_0^R r^2 \, dm$$

$$= \int_0^R r^2 \cdot \frac{2m}{R^2} \, r \, dr$$

$$= \frac{2m}{R^2} \int_0^R r^3 dr$$

$$= \frac{2m}{R^2} \left[\frac{r^4}{4} \right]_0^R$$

$$= \frac{2m}{R^2} \cdot \frac{R^4}{4}$$

$$I_{zz} = \frac{mR^2}{2} \quad [kg \cdot m^2]$$

Cheat Sheet

About the Center of Mass, across the rod : $dm = \frac{M}{L} dr$, $\int_{-L/2}^{L/2}$ $\Rightarrow I = \frac{ML^2}{12}$

About the pivot (end), across the rod : $dm = \frac{M}{L} dr$, $\int_0^L$ $\Rightarrow I = \frac{ML^2}{3}$

About the long axis : $dm = \frac{2m}{R^2} r \, dr$, $\int_0^R$ $\Rightarrow I = \frac{mR^2}{2}$

Torque

Torque [N·m] is how strongly a push tries to twist an object about an axis through the pivot

↳ the quantity that changes a spin, the way force is the quantity that changes straight-line motion

$F = ma$ ⟷ $\tau = I\alpha$

Force = mass × acceleration

[N] [kg] [m/s²]

Torque = Inertia × angular acceleration

[N·m] [kg·m²] $\frac{d\omega}{dt} = \left[\frac{1}{s^2}\right]$

↳ Why multiply N"·m"?

$\tau = F_\perp \cdot d$ where d is the perpendicular distance from the pivot to the force line and for rotation, where we apply force matters not only how big it is

- Angular → Tangential

↳ Converts a rotational quantity to linear quantity

∵ even same angle movement, the arc length differs by the distance r

$S = r \cdot \theta \Rightarrow [m] = [m] \cdot [rad]$

- Why $\tau = r \cdot F_\perp$?

0) we derived $\tau = I\alpha$

1) NL2 : $F_t = m\alpha_t$ → tangential acceleration $[m/s^2]$
∵ NL2 is linear motion

Tangential Angular

2) arc length : $s = r\theta$

3) differentiate w.r.t time : (r is constant)

$$\frac{ds}{dt} = r\frac{d\theta}{dt}$$

$$v_t = r\omega$$

tangential velocity, angular velocity

4) differentiate w.r.t time :

$$\frac{dv_t}{dt} = r\frac{d\omega}{dt}$$

$$\alpha_t = r\alpha$$

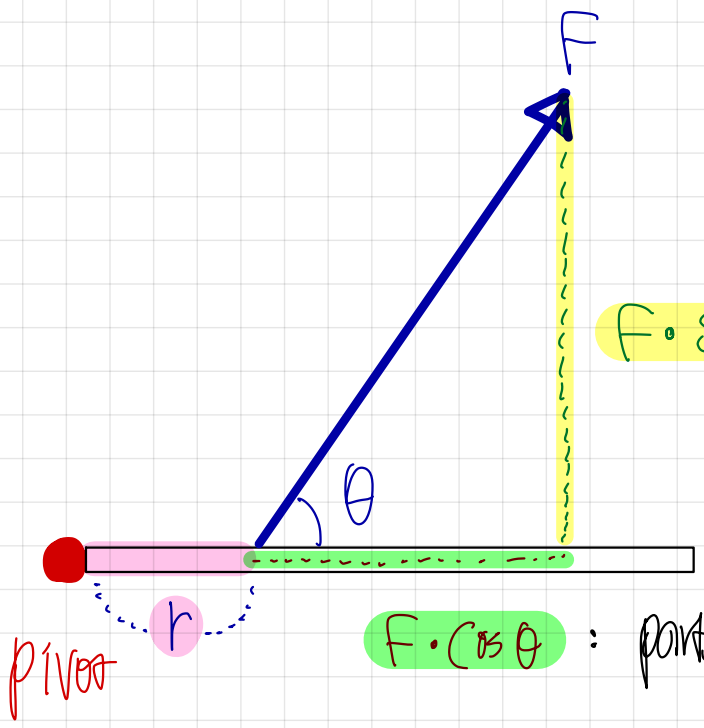
tangential acceleration, angular acceleration

5) combine and multiply r : $F_\perp = m(r\alpha)$

$$rF_\perp = mr^2\alpha = I\alpha = \tau$$

($mr^2 = I$)

- Push at an angle



$F \cdot \sin\theta$: only parts that matters for the spin

$F \cdot \cos\theta$: parts that does Nothing for the spin

∴ $\tau = r \cdot F \sin\theta$

↳ $\theta = 90^\circ$ gives the max rF as $\sin\theta = 1$

- Sign of τ

i) Counterclockwise : +

ii) Clockwise : −

- Describe both arrows by length and angle (polar form)

↳ ρ [m] : the length of the position vector

φ_r : angle from ρ axis

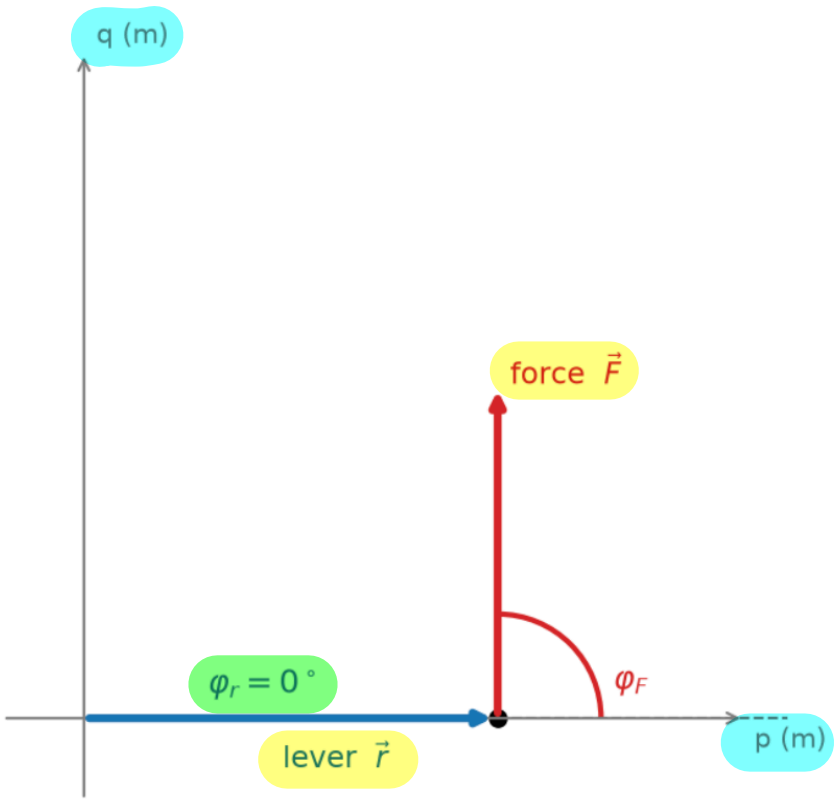
f [N] : the force's length

φ_F : angle from ρ axis

Two arrows on one flat page: lever (blue) and force (red)

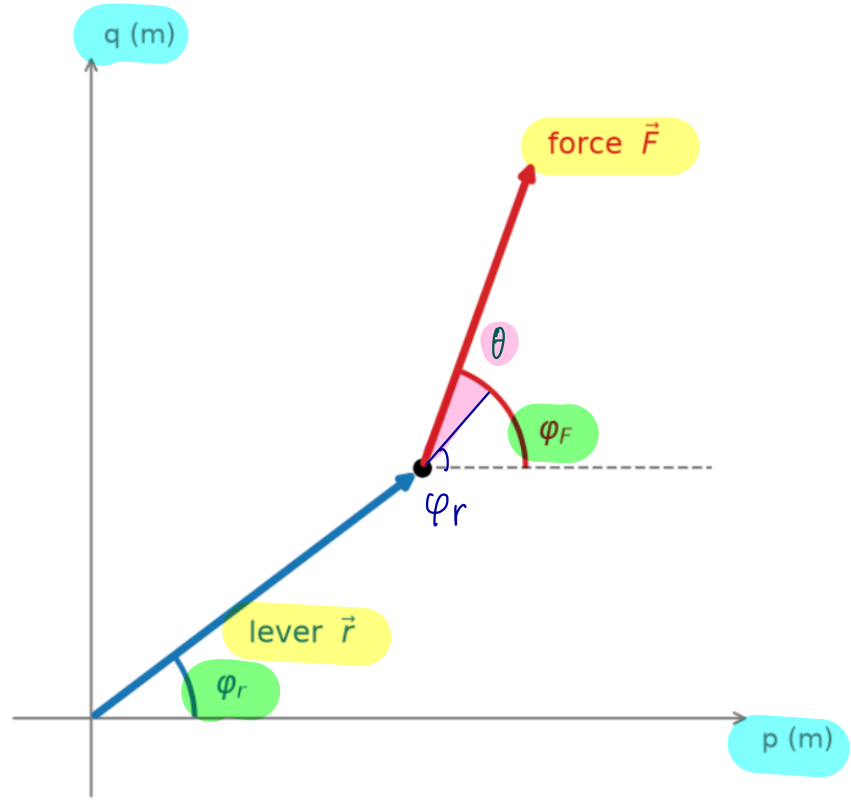
Rod to (1, 0) m, push up

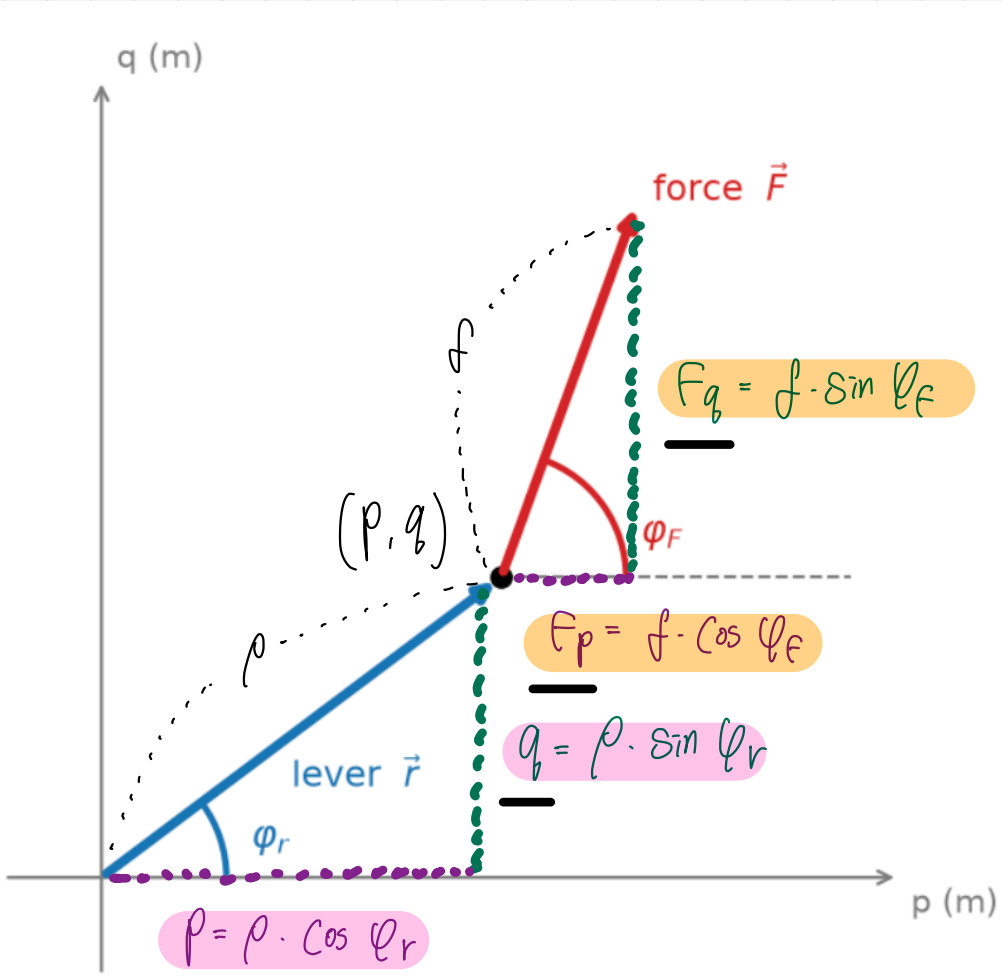
$\rho = 1.00$ m, $\varphi_r = 0.0^\circ$
$f = 10.0$ N, $\varphi_F = 90.0^\circ$
$\theta = \varphi_F - \varphi_r = 90.0^\circ$
$\tau = pF_q - qF_p = 10.00$ N·m



General case: both angles nonzero

$\rho = 1.00$ m, $\varphi_r = 36.9^\circ$
$f = 10.0$ N, $\varphi_F = 70.0^\circ$
$\theta = \varphi_F - \varphi_r = 33.1^\circ$
$\tau = pF_q - qF_p = 5.47$ N·m





$\theta = \varphi_F - \varphi_r$ (w.r.t φ_r, how much φ_F moves in angles?)

Positive θ means the force points counterclockwise from the lever

So the spin is counterclockwise

$$\therefore \quad I = r \cdot F$$

$$= \rho \cdot f \sin(\varphi_F - \varphi_r)$$

↳ carries the magnitude and the sign of direction at once

↳ we know $\sin(A-B) = \sin A \cos B - \cos A \sin B$

since $\cos(A-B) = \cos A \cos B + \sin A \sin B$

$$\text{so} \quad I = \rho \cdot f(\sin\varphi_F \cos\varphi_r - \cos\varphi_F \sin\varphi_r)$$

$$= (\rho\cos\varphi_r)(f\sin\varphi_F) - (\rho\sin\varphi_r)(f\cos\varphi_F)$$

$$\therefore \quad I = pF_q - qF_p$$

- Spin : described by 3 things
  - i) axis (the line everything turns around)
  - ii) sense (which way round)
  - iii) rate ω [rad/s]

$\Rightarrow$ We want to write all in one arrow $\vec{\omega}$ lying along the axis

- Arc length = radius $\times$ angle moved in rad

$S = r\theta$ $\quad$ r : a point's distance from the axis

so $r\theta$ = arc length

$$\text{L} \quad v = \frac{d(r\theta)}{dt} = r\frac{d\theta}{dt} = r\omega \quad [m/s]$$

- Angular velocity $\omega = \frac{d\theta}{dt}$ $[1/s]$

L rate of change in angle

Angular velocity vector

- Angular velocity vector $\vec{\omega}$ has

i) the direction along the axis of rotation with the end chosen by the right-hand rule

ii) the magnitude equal to the angular speed $|\omega| = \left|\frac{d\theta}{dt}\right|$

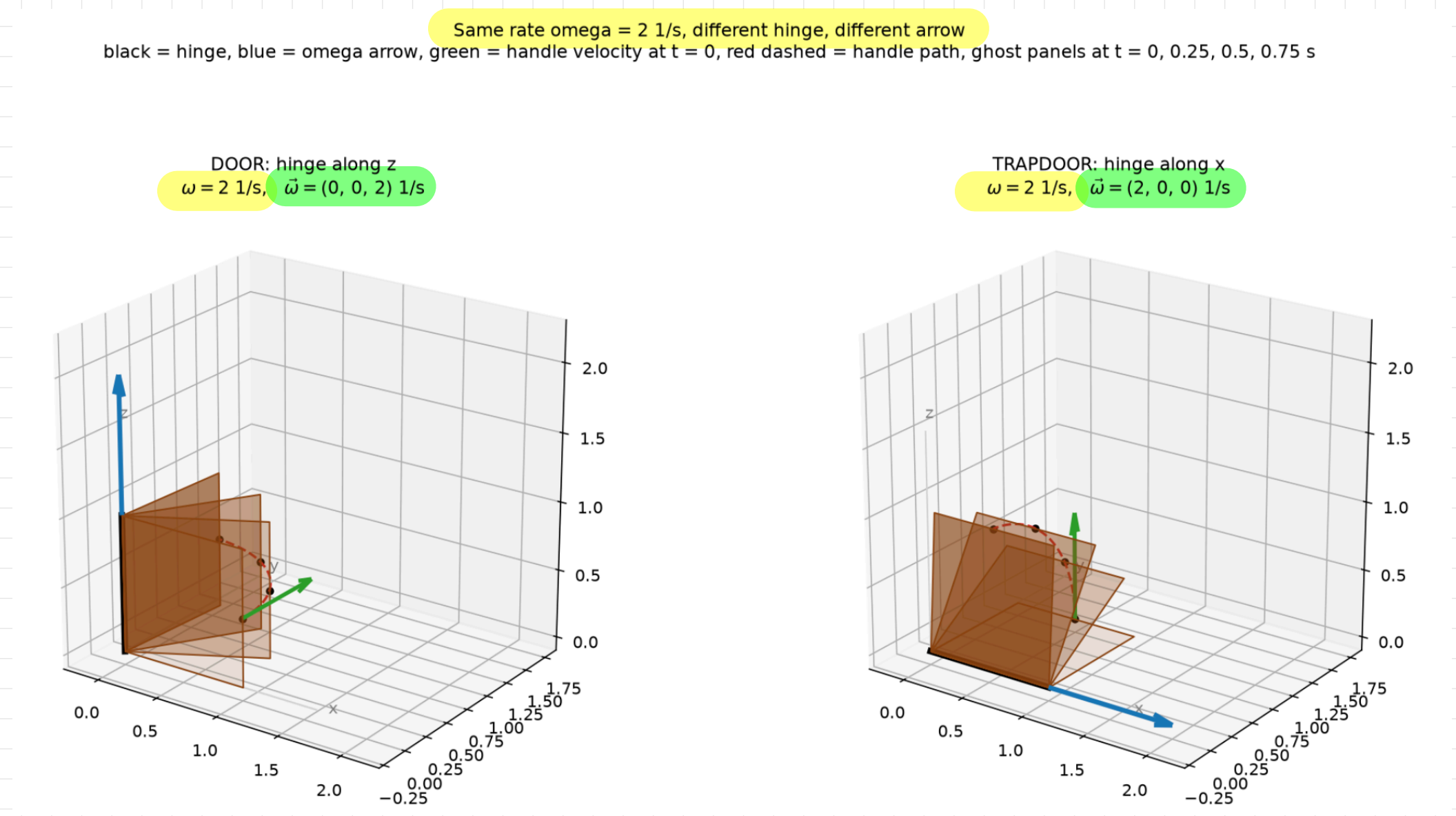
L $\vec{\omega} = \omega \hat{n}$

= rate attaches to the axis of the turning

where $\hat{n}$ is the unit arrow along the axis

↳ $\hat{x} = (1,0,0)$, $\hat{y} = (0,1,0)$, $\hat{z} = (0,0,1)$

= we need $\vec{\omega}$ not only ω since in 3D same ω can belong to different turning



↳ unit arrow $\hat{n} = (n_x, n_y, n_z)$

$$|\hat{n}| = \sqrt{n_x^2 + n_y^2 + n_z^2} = 1$$

ex) $(1, 0, 0)$

$(0.\text{non}1, 0.\text{non}1, 0)$

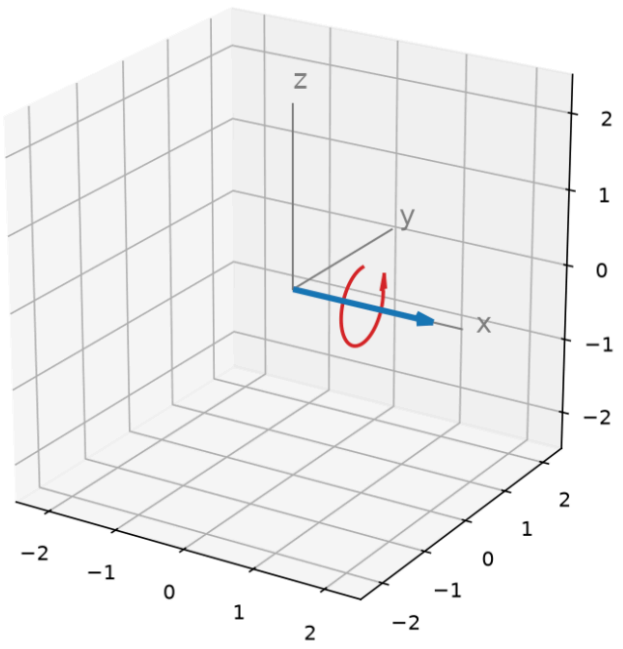
$(0.5\text{nn}4, 0.5\text{nn}4, 0.5\text{nn}4)$

$$\vec{\omega} = (\omega_x, \omega_y, \omega_z) \qquad \left[\frac{\text{rad}}{\text{s}}\right]$$

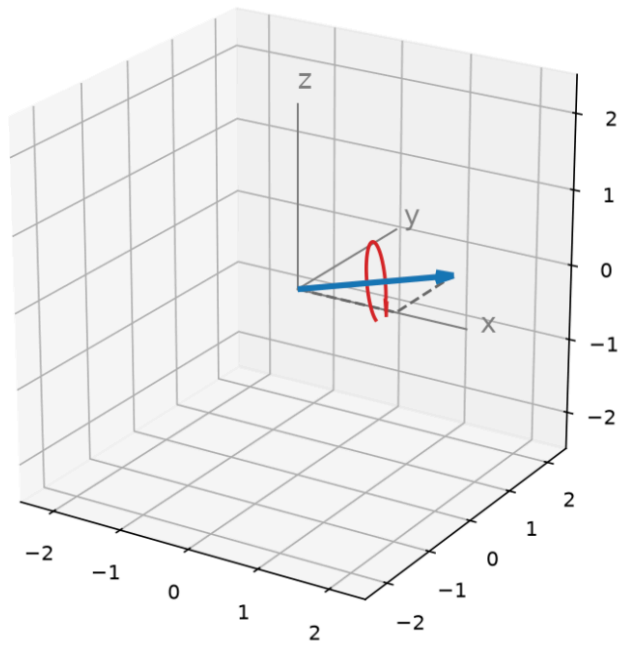
$\Rightarrow$ ω_N : the part of the turning that is about the N axis

Same rate 2 1/s, three unit axes: blue = omega arrow, red = sense, dashed = slots added one after another

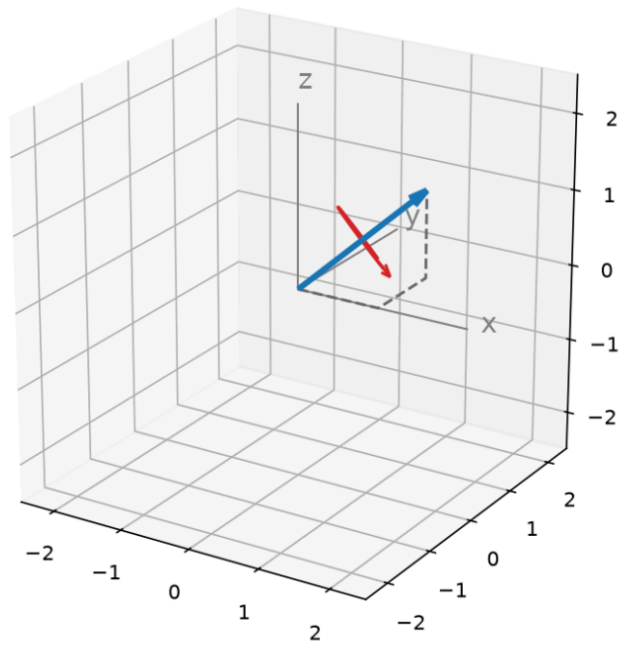
$\hat{n} = (1.0000,\ 0.0000,\ 0.0000)$, w=2 1/s
longitude 0.0°, latitude 0.0°
$\vec{\omega} = (2.000,\ 0.000,\ 0.000)$ 1/s, $|\vec{\omega}| = 2.000$ 1/s



$\hat{n} = (0.7071,\ 0.7071,\ 0.0000)$, w=2 1/s
longitude 45.0°, latitude 0.0°
$\vec{\omega} = (1.414,\ 1.414,\ 0.000)$ 1/s, $|\vec{\omega}| = 2.000$ 1/s

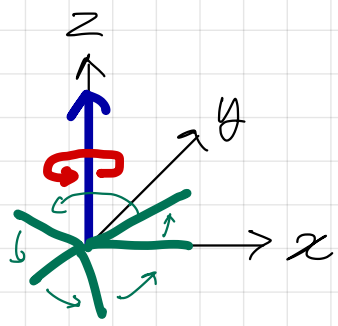


$\hat{n} = (0.5774,\ 0.5774,\ 0.5774)$, w=2 1/s
longitude 45.0°, latitude 35.3°
$\vec{\omega} = (1.155,\ 1.155,\ 1.155)$ 1/s, $|\vec{\omega}| = 2.000$ 1/s



In 2D case, we are stuck at using $\hat{n} = (0, 0, 1)$

so the only choice is $\vec{\omega} = (0, 0, \omega_z)$ or $(0, 0, -1)$



- What the $\vec{\omega}$ means

ω answers "how fast overall"

$\hat{n}$ answers "around which line"

$\vec{\omega}$ answers both at once, by splitting the overall turning into shares about x, y, and z

$$\omega = \|\vec{\omega}\| = \sqrt{\omega_x^2 + \omega_y^2 + \omega_z^2}$$

$$\vec{\omega} = (\omega \cos\theta_x,\ \omega \cos\theta_y,\ \omega \cos\theta_z)$$

- $\vec{\omega} = (1.414, 1.414, 0)$ and Interpretation

  - Both x and y are involved in turning by 1.414 $^{g}/_{s}$

$\vec{\omega}$ = 1.414 $^{g}/_{s}$ about x + 1.414 $^{g}/_{s}$ about y + 0 $^{g}/_{s}$ about z

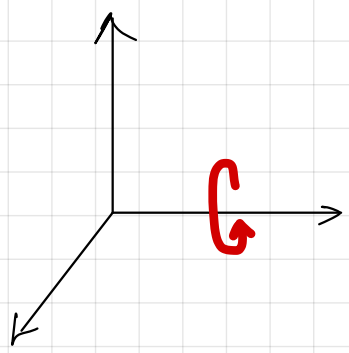
ω_x = $2 \times \cos 45^\circ$

ω_y = $2 \times \cos 45^\circ$

ω_z = $2 \times \cos 90^\circ$

  - turning of 1 $^{g}/_{s}$ about x:

    the body spins around the x axis



  - Interpretation) A turning at 2 $^{g}/_{s}$ about an axis halfway between x and y is the same as a turning at 1.414 $^{g}/_{s}$ about x and a turning at 1.414 $^{g}/_{s}$ about y happening together

Right Hand Rule

Angular Acceleration

- $\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \left(\frac{d\omega_x}{dt}, \frac{d\omega_y}{dt}, \frac{d\omega_z}{dt} \right)$ $\left[{}^{1}/_{s^2} \right]$

  ↳ Since in 2D, $\alpha = \frac{d\omega}{dt}$

- $\vec{\tau} = \tau \hat{n}$: the torque arrow points along the axis the push is trying to turn the body about, with the end chosen by the right hand, and its length τ is the strength of the twist

  ↳ For a simple setup (a point mass m on a massless rod, turning about an axis perpendicular to the rod), we have $\tau = I\alpha$

  $\Rightarrow \vec{\tau} = I\vec{\alpha}$

- 3D torque arrows from a lever $\vec{r}$ and a push $\vec{F}$

  ↳ we derived $\tau = pF_q - qF_p$

  ↳ Choose the order of p, q so the counterclockwise matches the right-hand thumb along the axis

Axis	Plane order (p, q)
z	(x, y)
x	$(0, z)$
y	(z, x)

$\therefore$ $I_z = r_x f_y - r_y f_x$ (xy plane)

$I_x = r_y f_z - r_z f_y$ (yz plane)

$I_y = r_z f_x - r_x f_z$ (zx plane)

$\Rightarrow$ $\vec{I} = (I_x, I_y, I_z)$

Dot product

Cross Product

(a_x, a_y, a_z) (b_x, b_y, b_z)

$a \times b$ of two vectors a and b is a third vector that

i) is perpendicular to both and

ii) has a length $|a||b| \sin\theta$ where θ is an angle betw

iii) points in the direction given by the right-hand rule

Where the thumb points

↳ Order matters: a is the first and b the second

$$a \times b = -b \times a$$

Tangential Velocity in 3D

The tangential velocity of a point on a rigid body spinning about an axis is the velocity of that point

$$\vec{v} = \vec{\omega} \times \vec{r} \qquad [m/s] = [1/s]\,[m]$$

cross
- $\vec{r}$ → position of the point measured from the origin O
- $\vec{\omega}$ → the spin arrow

- we have

i) Position $\vec{r}$ and velocity $\vec{v}$ always

$$\vec{r} = (x, y, z) \quad , \quad \vec{v} = \frac{d\vec{r}}{dt} = (\dot{x}, \dot{y}, \dot{z})$$

ii) Tangential velocity at 2D

$$\vec{r} = \rho(\cos\theta, \sin\theta) \quad , \quad \vec{v} = \rho\omega(-\sin\theta, \cos\theta)$$

→ $\omega = \dot{\theta}$ $[1/s]$ the rate

iii) Angular velocity $\vec{\omega}$

$$\vec{\omega} = \omega\hat{n}$$

iv) Unit arrow $\hat{n}$

$(X \to Y \to Z \to X)$

$\hat{z} \times \hat{x} = \hat{y}$, $\hat{z} \times \hat{y} = -\hat{x}$, $\hat{z} \times \hat{z} = 0$

v) Cross Product

$\vec{a} \times \vec{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_z)$

and at right angle and both $\vec{a}$ and $\vec{b}$ with size $|a||b| \sin\varphi$

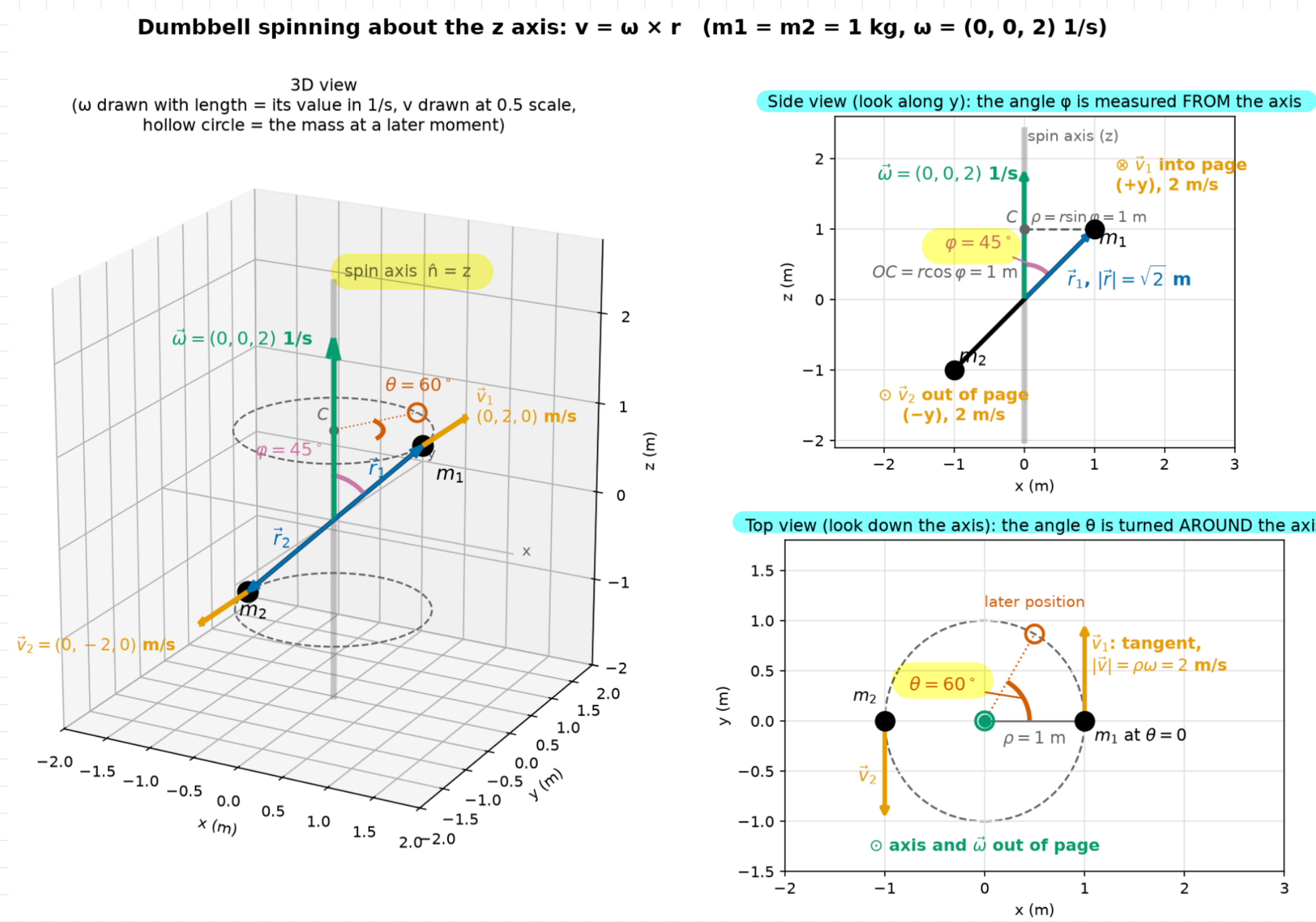
↳ What does spinning about an axis mean?)

Every point of a rigid body

1) keeps its position along the axis

2) keeps its distance ρ from the axis (ρ : angle from the side)

3) moves around the axis by the angle θ (θ : angle from the top)



↳ proof) With $\vec{w}$ and $\vec{r}$ is perpendicular to $\vec{v}$?

Why use a Cross product ?

① Case 1 : The Spin axis is Z axis

i) $\hat{n} = \hat{z} \Rightarrow \vec{\omega} = (0, 0, \omega_z) = (0, 0, \omega)$

ii) Set θ_0 where the point Stores (so Starting angle)
and $\theta(t)$ what the point has turned Since $t = 0$
z_0 the Starting height

Position Vector at time t after the body has turned $\theta(t)$)

$\vec{r}(t) = (\rho \cos(\theta_0 + \theta), \rho \sin(\theta_0 + \theta), z_0)$

iii) Differentiate to get the $\vec{V}(t)$

$$\vec{V}(t) = (-\rho\omega\sin(\theta_0+\theta),\ \rho\omega\cos(\theta_0+\theta),\ 0)$$

$$\vec{V}(t) = \omega(-y,\ x,\ 0)$$

iv) Compare $\vec{V}(t)$ with the cross product of $\vec{\omega}$ and $\vec{r}$

Same!

$$\vec{\omega}\times\vec{r} = (0\cdot z - \omega y,\ \omega x - 0\cdot z,\ 0)$$

$$\vec{\omega}\times\vec{r} = \omega(-y,\ x,\ 0)$$

∴ For the spin in Z-axis, $\vec{V} = \vec{\omega}\times\vec{r}$

② Any Axis

1) Split $\vec{r} = \vec{r}_{\parallel} + \vec{r}_{\perp}$:

$\vec{r}_{\parallel} = (\vec{r}\cdot\hat{n})\,\hat{n}$ → Since $\vec{r}\cdot\vec{n}$ is constant

$\vec{r}_{\parallel}$ is a constant vector

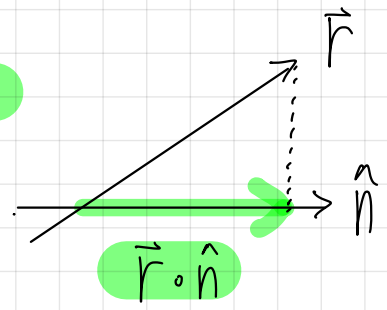
$\vec{r}_{\perp} = \vec{r} - \vec{r}_{\parallel}$

$\rho = |\vec{r}_{\perp}| = r\sin\theta$

↳ $\vec{r}\cdot\hat{n}$: how far the $\hat{r}$ sits along the axis $\hat{n}$ (the length along the axis)

Dot product $\vec{r}\cdot\hat{n} = |r||n|\cos\theta = r\cos\theta$

↳ $|\hat{n}| = 1$ since unit



↳ $\vec{r}_\perp \cdot \hat{n}$ where $\vec{r}_\perp = \vec{r} - \vec{r}_{||}$ ∴

$$\vec{r}_\perp \cdot \hat{n} = (\vec{r} - \vec{r}_{||})\,\hat{n} = \vec{r}\cdot\hat{n} - (\vec{r}\cdot\hat{n})\underbrace{(\hat{n}\cdot\hat{n})}_{1} = 0$$

↳ $\rho = |\vec{r}_\perp| = r\sin\varphi$ ∴

$$r^2 = |\vec{r}_{||}| + \rho^2 \Rightarrow \rho^2 = r^2 - |\vec{r}_{||}|$$

$$= r^2 - r^2\cos^2\varphi$$

$$= r^2\sin^2\varphi$$

$$\therefore \rho = r\sin\varphi$$

2) Build a frame that plays the role of $(\hat{x}, \hat{y}, \hat{z})$ ∴

$e_2 = \dfrac{\vec{r}_\perp(0)}{\rho}$, $e_2 = \hat{n} \times e_1$, $e_3 = \hat{n}$

they are unit arrow at right angles to each other

$$\hat{n} \times e_1 = e_2 \,, \quad \hat{n} \times e_2 = -e_1 \,, \quad \hat{n} \times \hat{n} = 0$$

↳ e_1) Since $\vec{r}_{||}$ is whatever part of $\vec{r}$ that is parallel to $\hat{n}$

so $\vec{r}_\perp$ is perpendicular to $\hat{n}$, then normalized with ρ

3) Position at time t :

$$\vec{r}(t) = \vec{r}_{\parallel} + \rho(\cos\theta\, e_1 + \sin\theta\, e_2)$$

↳ Since e_1 and e_2 play the unit arrow roles

$$\vec{r}_{\perp} = a\, e_2 + b\, e_2$$

Since $|\vec{r}_{\perp}| = \rho$, we have $\rho^2 = a^2 + b^2$

let angle ψ be measured from e_1 towards e_2

$\Rightarrow$ $a = \rho\cos\psi$ $\quad$ $b = \rho\sin\psi$

4) Differentiate W.r.t time

$$\vec{v} = \rho\omega(-\sin\theta \cdot e_1 + \cos\theta \cdot e_2)$$

5) Compare $\vec{v}$ with the cross product of $\vec{\omega} = \omega\hat{n}$ and $\vec{r}$

↳ $\vec{\omega} \times \vec{r} = \omega(\hat{n} \times \vec{r}) = \omega\hat{n} \times (\vec{r}_{\parallel} + \rho\cos\theta e_1 + \rho\sin\theta e_2)$

$$= \omega \left[\left(\hat{n} \times \vec{r}_{\perp} \right) + \left(\hat{n} \times \rho \cos\theta \, e_1 \right) + \left(\hat{n} \times \rho \sin\theta \, e_1 \right) \right]$$

$$= \omega \left[\underbrace{\left(\hat{n} \times \vec{r}_{\perp} \right)}_{0} + \rho \cos\theta \underbrace{\left(\hat{n} \times e_1 \right)}_{e_2} + \rho \sin\theta \underbrace{\left(\hat{n} \times e_2 \right)}_{-e_1} \right]$$

$$\vec{\omega} \times \vec{r} = \rho \omega \left(-\sin\theta \, e_1 + \cos\theta \, e_2 \right)$$

∴ Cross product proves

Momentum

How hard is it to stop this object?

The amount of straight line motion a body has.

$$\vec{p} = m \cdot \vec{v} \qquad [kg \cdot m/s] = [kg] \cdot [m/s]$$

- Derivation)

For a constant mass,

$$F = ma = m\left(\frac{dv}{dt}\right) = \frac{d(mv)}{dt} = \frac{dp}{dt}$$

↓

amount of motion mv per unit time

Angular Momentum

$$\vec{L} = r m v_t = r \vec{p}_t = I \vec{\omega}$$

0) We derives $L = r F_t = r(m a_t)$

1) Since $a_t = \frac{dv_t}{dt}$ and m is constant

$$m\,dt = \frac{d(mV_t)}{dt} \quad \text{so} \quad I = r\,\frac{d(mV_t)}{dt}$$

a) $$I = \frac{d(r\,m\,V_t)}{dt} = \frac{d\,L}{dt}$$

↳ amount of angular motion rmV_t per unit time

4) $$L = r\,m\,V_t = r\,p_\perp \qquad [m \cdot kg \cdot m/s] = [kg \cdot m^2/s]$$

↳ where $p = mV_t$

(momentum across the rod)

5) $$L = r\,m(r\omega) = \underbrace{m\,r^2}_{I}\,\omega = I\,\omega$$

6) $$\frac{dL}{dt} = I\,\frac{d\omega}{dt} = I\,\alpha = I$$

$$\vec{L} = r\,\vec{p}_\perp = I\,\vec{\omega}$$

Inertia Tensor

A body can turn about three axes, and a torque about one axis can produce angular acceleration about another. We use 3x3 table for x, y, z axis.

$I_{\alpha\beta}$ = the torque about axis that α that we need, per unit of angular acceleration about axis β

Inertia Tensor (I) : belongs to the rigid body and it converts the body's angular velocity $\vec{\omega}$ [1/s] into its angular momentum $\vec{L}$ [kg · m²/s]

$$\vec{L} = I\,\vec{\omega}$$

(3x1) (3x3) (3x1)

Column = axis you spin about (ω : angular velocity)

	x	y	z
x	I_{xx}	I_{xy}	I_{xz}
y	I_{yx}	I_{yy}	I_{yz}
z	I_{zx}	I_{zy}	I_{zz}

row = Which component of L (or torque) you look at

Problem : In 2D, $L = I\omega$ with a single number I

In 3D, one number to transform ω to L is

Not enough

ex) Does single number I exists with $L = I\omega$ or $\vec{L} = I\vec{\omega}$?

2D : $L = mr^2\omega = I\omega$ with $I = mr^2$ ∴ exist(Y)

3D : $\vec{r_1} = (1,0,1)$, $\vec{r_2} = (-1,0,-1)$, $\omega = (0,0,\omega)$

	mass 1	mass 2
tangential velocity $\vec{v} = \vec{\omega} \times \vec{r}$	$(0,\omega,0)$	$(0,-\omega,0)$
$\vec{L_i} = \vec{r_i} \times m\vec{v_i}$	$(-\omega,0,\omega)$	$(-\omega,0,\omega)$

Formulas used (components): $\vec{r} = (x,y,z)$

- tangential velocity $\vec{v} = \vec{\omega} \times \vec{r} = (\omega_y z - \omega_z y,\ \omega_z x - \omega_x z,\ \omega_x y - \omega_y x)$
- angular momentum of one mass $\vec{L} = \vec{r} \times m\vec{v} = m\,(y v_z - z v_y,\ z v_x - x v_z,\ x v_y - y v_x)$

Mass 1, $\vec{r}_1 = (1, 0, 1)$:

$$\vec{v}_1 = (0 \cdot 1 - \omega \cdot 0,\ \omega \cdot 1 - 0 \cdot 1,\ 0 \cdot 0 - 0 \cdot 1) = (0,\ \omega,\ 0)$$

$$\vec{L}_1 = (0 \cdot 0 - 1 \cdot \omega,\ 1 \cdot 0 - 1 \cdot 0,\ 1 \cdot \omega - 0 \cdot 0) = (-\omega,\ 0,\ \omega)$$

Mass 2, $\vec{r}_2 = (-1, 0, -1)$:

$$\vec{v}_2 = (0 \cdot (-1) - \omega \cdot 0,\ \omega \cdot (-1) - 0,\ 0) = (0,\ -\omega,\ 0)$$

$$\vec{L}_2 = (0 \cdot 0 - (-1)(-\omega),\ (-1) \cdot 0 - (-1) \cdot 0,\ (-1)(-\omega) - 0) = (-\omega,\ 0,\ \omega)$$

Total: $\vec{L} = \vec{L}_1 + \vec{L}_2 = (-2\omega,\ 0,\ 2\omega)$ kg·m²/s.

↳ So the rotation only happens in Z-axis but $\vec{L}$ has x components. so $\vec{L}$ is not parallel to $\vec{\omega}$, I can only rescale (so one number cannot link them)

∴ Cannot use a single number I to express $\vec{L}$ (X)

⇒ We want to find how $\vec{L}$ and $\vec{\omega}$ are related

- $\vec{L}$ depends on $\vec{\omega}$ linearly

↳ we know $L = r\,m(v_t) = r\,m(r\omega)$

↳ Rigid Body : Every mass shares the same $\vec{\omega}$ so $\vec{v}_i = \vec{\omega} \times \vec{r}_i$

↳ m_i and $\vec{r}_i$ are fixed numbers that does not depends on $\vec{\omega}$

↳ Linear means "Scalar Multiplication" and "Additivity" laws are met

⇒ Therefore) $\left(L = r\,m\,r_t = r\,m(r\omega) = (r^2 m)\omega = I\omega \right)$

for one mass i, $\vec{L}_i(\vec{\omega}) = m_i \vec{r}_i \times (\vec{\omega} \times \vec{r}_i)$

So for all masses, the body's angular momentum vector is

$$\vec{L} = \sum_i \vec{L}_i$$

i) Scalar Multiplication : $\vec{\omega} \to k\vec{\omega}$

$$\vec{r_i} \times ((k\vec{\omega}) \times \vec{r_i}) = \vec{r_i} \times k(\vec{\omega} \times \vec{r_i}) = k(\vec{r_i} \times (\vec{\omega} \times \vec{r_i})) \quad //$$

ii) Additivity : $\vec{\omega} \to \vec{\omega_1} + \vec{\omega_2}$

$$\vec{r_i} \times ((\vec{\omega_1} + \vec{\omega_2}) \times \vec{r_i}) = \vec{r_i} \times ((\vec{\omega_1} \times \vec{r_i}) + (\vec{\omega_2} \times \vec{r_i}))$$

$$= \vec{r_i}(\vec{\omega_1} \times \vec{r_i}) + \vec{r_i}(\vec{\omega_2} \times \vec{r_i})$$

$$\vec{L_i}(\vec{\omega_1} + \vec{\omega_2}) = \vec{L_i}(\vec{\omega_1}) + \vec{L_i}(\vec{\omega_2}) \quad //$$

$\Rightarrow$ Sum of Linear rules is Linear so the whole body $\vec{L}$ is Linear in $\vec{\omega}$

L. Why?) $\vec{\omega}$ appears only once in $\vec{L}$ never squared unlike kinetic energy $\frac{I\omega^2}{2}$ (quadratic)

Test	Calculation	Result
$\vec{\omega} = (0,0,1)$	$\vec{v} = (0,1,0), \vec{L}_1 = (1,0,1) \times (0,1,0)$	$(-1,\ 0,\ 1)$
Scaling: $\vec{\omega} = (0,0,3)$	$\vec{v} = (0,3,0), \vec{L}_1 = (1,0,1) \times (0,3,0)$	$(-3,\ 0,\ 3) = 3 \cdot (-1,0,1)$ ✔
$\vec{\omega} = (1,0,0)$	$\vec{v} = (0,-1,0), \vec{L}_1 = (1,0,1) \times (0,-1,0)$	$(1,\ 0,\ -1)$
Additivity: $\vec{\omega} = (1,0,1)$	$\vec{v} = (1,0,1) \times (1,0,1) = \vec{0}$, so $\vec{L}_1$	$(0,0,0) = (-1,0,1) + (1,0,-1)$ ✔

- Linear means Matrix

1) Any angular velocity vector is a sum of its parts along the unit vector

$$\vec{\omega} = \omega_x \hat{x} + \omega_y \hat{y} + \omega_z \hat{z}$$

2) Linearity's Additivity splits $\vec{L}(\vec{\omega})$ into three terms and Scalar Multiplication takes the numbers out $(L = rmv_t = rm(r\omega) = (r^2 m)\omega = I\omega)$

$$\vec{L}(\vec{\omega}) = \vec{L}(\omega_x \hat{x}) + \vec{L}(\omega_y \hat{y}) + \vec{L}(\omega_z \hat{z})$$

$$\vec{L}(\vec{\omega}) = \omega_x \vec{L}(\hat{x}) + \omega_y \vec{L}(\hat{y}) + \omega_z \vec{L}(\hat{z})$$

↳ $\vec{L}(\hat{b})$ means the angular momentum vector for a unit angular velocity about axis b (the I part from $L = I''I''\omega$)

3) Write three angular momentum vectors with I_{ab} formats

$$\vec{L}(\hat{x}) = \begin{pmatrix} I_{xx} \\ I_{yx} \\ I_{zx} \end{pmatrix} \quad \vec{L}(\hat{y}) = \begin{pmatrix} I_{xy} \\ I_{yy} \\ I_{zy} \end{pmatrix} \quad \vec{L}(\hat{z}) = \begin{pmatrix} I_{xz} \\ I_{yz} \\ I_{zz} \end{pmatrix}$$

↳ I_{ab} means the a component of $\vec{L}(\hat{b})$

So $L_a = I_{ax}\omega_x + I_{ay}\omega_y + I_{az}\omega_z \quad (a = x, y, z)$

$$\therefore \vec{L} = \begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_x \\ \omega_y \\ \omega_z \end{pmatrix}$$

$$I \qquad \vec{\omega}$$

↳ $\vec{L}$: One vector (an arrow in 3D) with 3 components

$$\vec{L} = L_x \hat{x} + L_y \hat{y} + L_z \hat{z}$$

↳ L_a (where $a = x, y,$ or z) : the component of $\vec{L}$ along axis a

how much the arrow is points that way

$\Rightarrow$ Does Not mean "the rotation happens about axis a"

$I\omega = L$

- ω: angular velocity (input) $[1/s]$
- I: matrix tobe $[kg \cdot m^2]$
- L: angular Momentum (output) $[kg \cdot m^2/s]$

$I\omega = L$

$(Ax = b)$

- naming rule of I : $I_{\text{output, input}}$

$$L = \begin{bmatrix} L_x \\ L_y \\ L_z \end{bmatrix} \rightarrow L_a = I_{ax}\omega_x + I_{ay}x_y + I_{az}x_z \quad \in x,y,z$$

- I_{ab} : per $1/s$ of spin (angular velocity) about axis b, how much angular Momentem is shown along axis a

- I is built from the masses (m) and position (x,y,z) alone. They are Not influenced by the inputs (ω). It has the same template for any set of point masses

- Build I matrix template

↳ One mass m at $\vec{r} = (x, y, z)$ with a general spin

$\vec{\omega} = (\omega_x, \omega_y, \omega_z)$

↳ $\vec{L} = \vec{r} \times m\vec{v} = \vec{r} \times m(\vec{\omega} \times \vec{r})$

↳ Cross product: $\vec{a} \times \vec{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x)$

1) $\vec{v} = \vec{\omega} \times \vec{r} = (\underbrace{\omega_y z - \omega_z y}_{v_x},\ \underbrace{\omega_z x - \omega_x z}_{v_y},\ \underbrace{\omega_x y - \omega_y x}_{v_z})$

2) $\vec{L} = \vec{r} \times m\vec{v} = m(\vec{r} \times \vec{v}) = \begin{pmatrix} L_x \\ L_y \\ L_z \end{pmatrix} = I\omega$

↳ $\vec{r} \times \vec{v} = (\underbrace{y v_z - z v_y}_{L_x/m},\ \underbrace{z v_x - x v_z}_{L_y/m},\ \underbrace{x v_y - y v_x}_{L_z/m})$

1) $L_x = m(y v_z - z v_y)$

$= m[y(\omega_x y - \omega_y x) - z(\omega_z x - \omega_x z)]$

$= m[\underbrace{(y^2 + z^2)}_{I_{xx}}\omega_x + \underbrace{(-xy)}_{I_{xy}}\omega_y + \underbrace{(-xz)}_{I_{xz}}\omega_z]$

2) $L_y = m(z v_x - x v_z)$

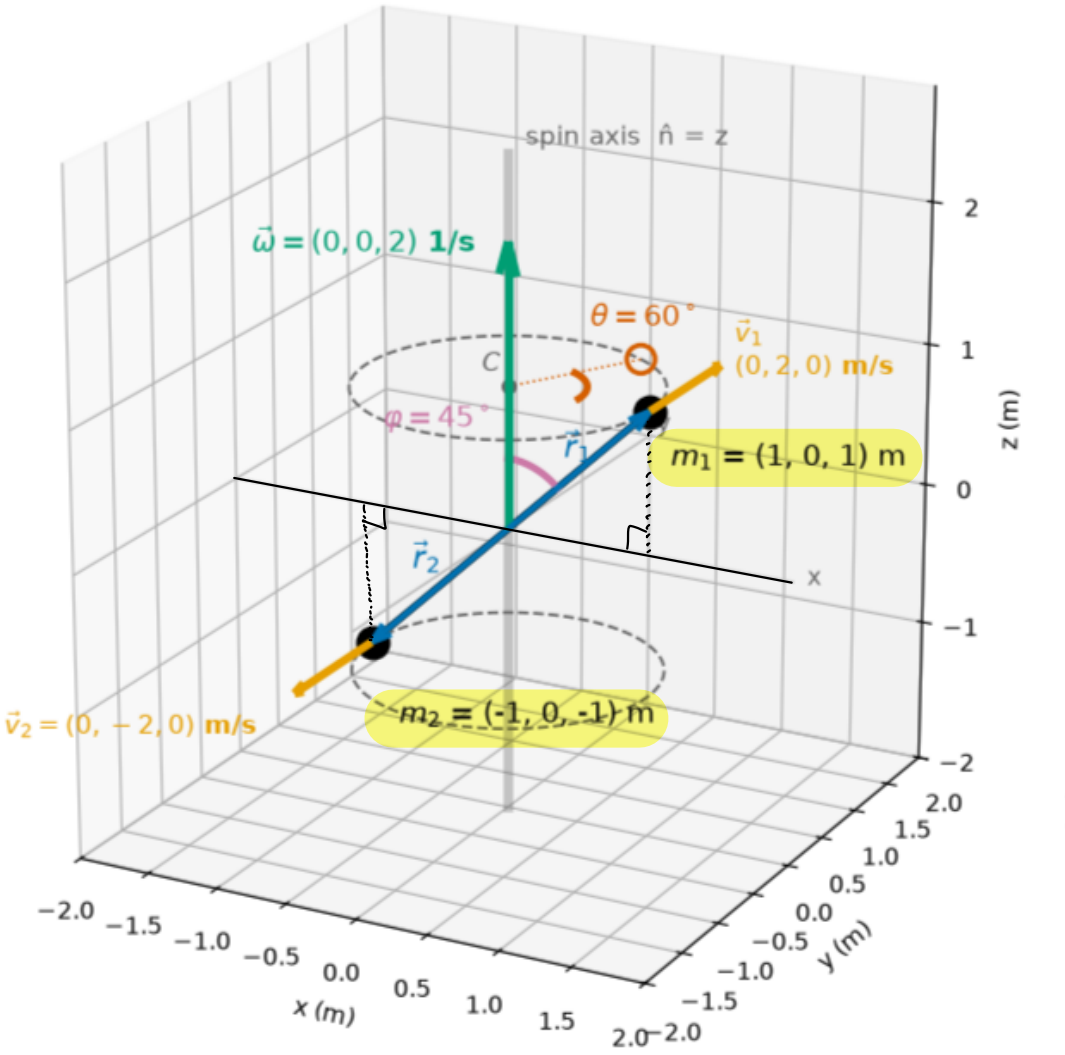
$$= m\left[\underbrace{(-xy)}_{I_{yx}}\omega_x + \underbrace{(x^2+z^2)}_{I_{yy}}\omega_y + \underbrace{(-yz)}_{I_{yz}}\omega_z\right]$$

3) $L_z = m(x v_y - y v_x)$

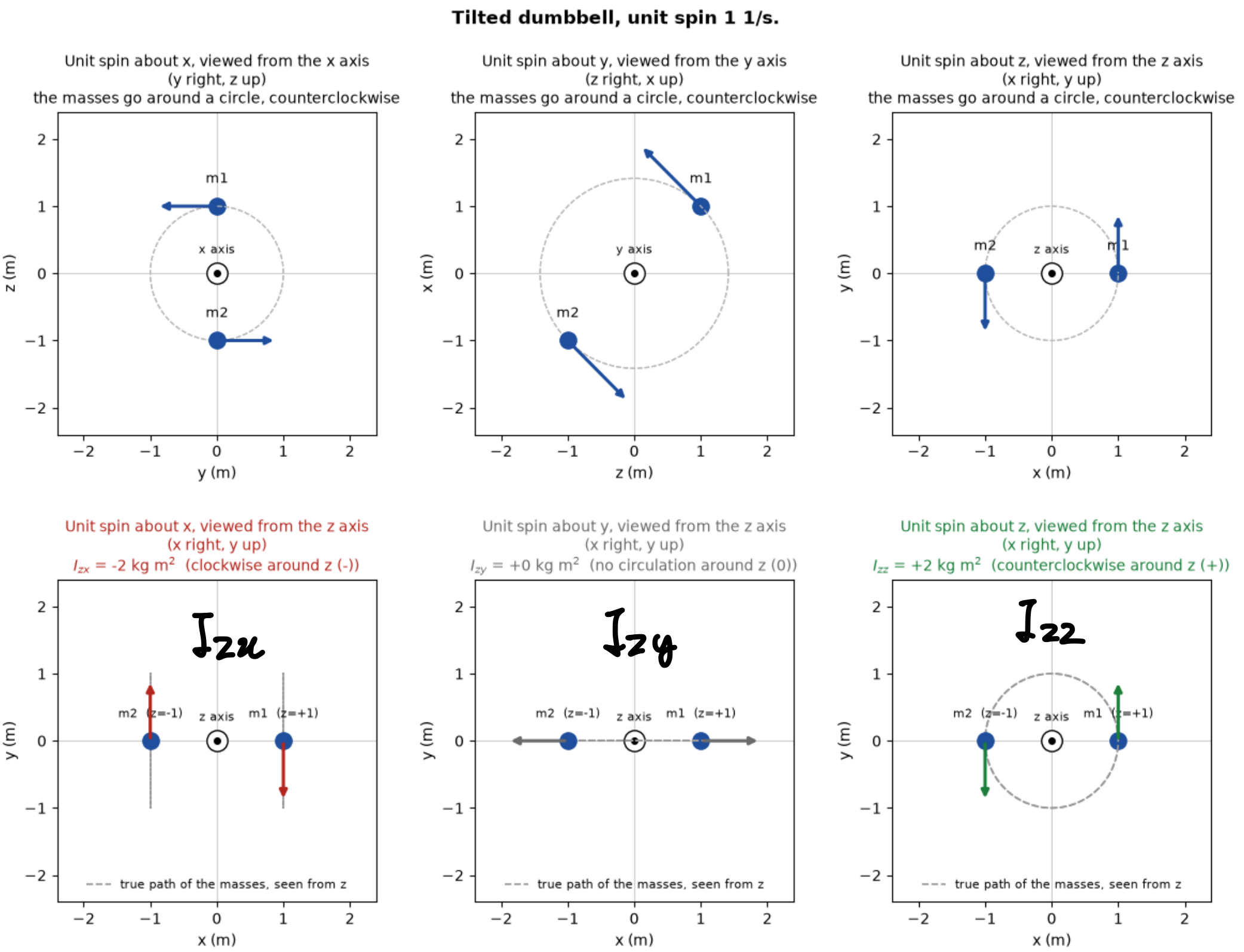
$$= m\left[\underbrace{(-xz)}_{I_{zx}}\omega_x + \underbrace{(-yz)}_{I_{zy}}\omega_y + \underbrace{(x^2+y^2)}_{I_{zz}}\omega_z\right]$$

3)

$$I = m\begin{bmatrix} y^2+z^2 & -xy & -zx \\ -xy & z^2+x^2 & -yz \\ -zx & -yz & x^2+y^2 \end{bmatrix}$$



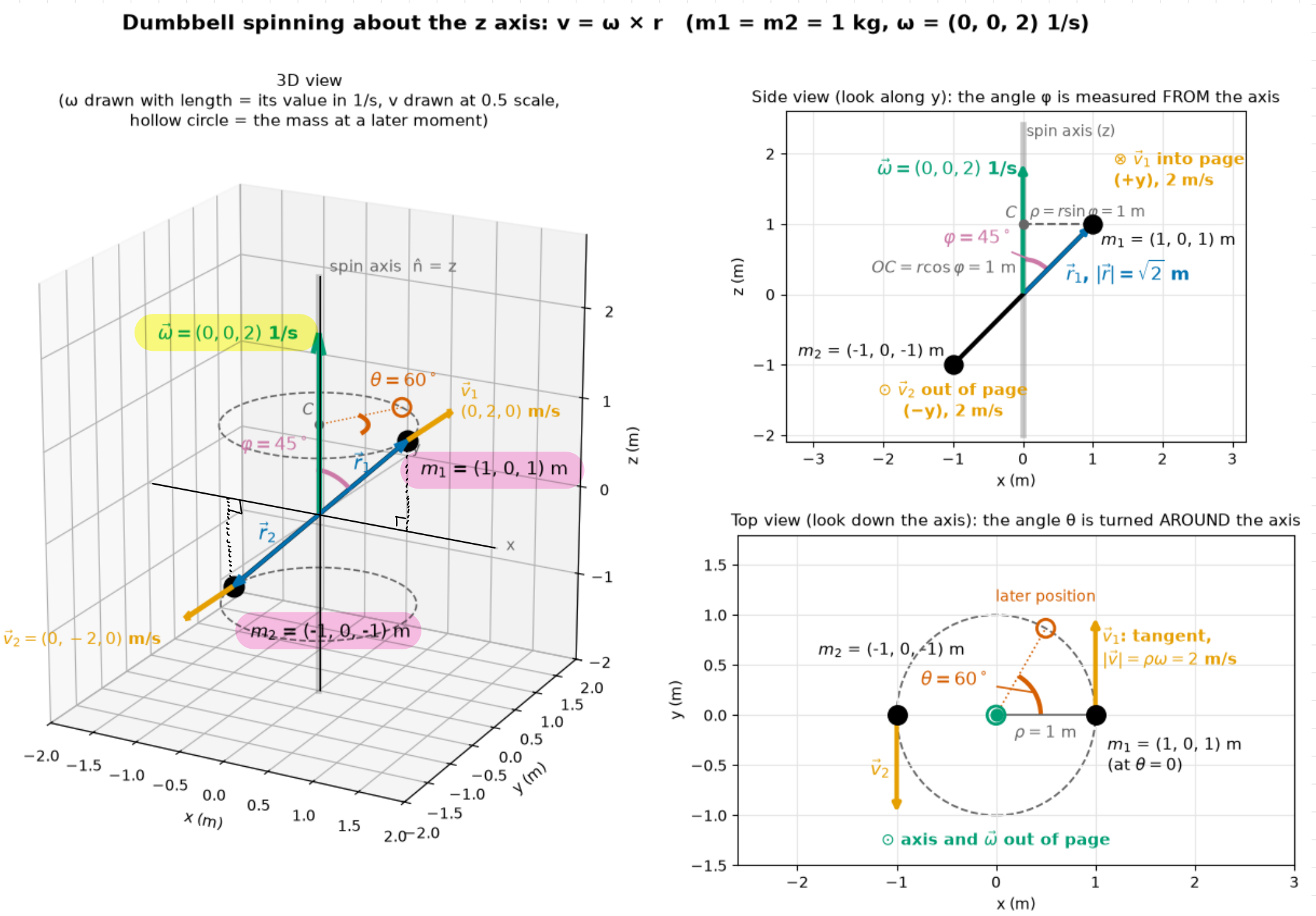
Interpret I_{ab}



- I_{zx} = a unit spin ($1\frac{1}{s}$) about the x-axis. View the spin from the z-axis and read z-component of $\vec{L}$ (circulation around z)

↳ Diagonal entries (a=b) are the cases where the view and the spin share the axis

Dumbbell Problem



$$\vec{\omega} = \vec{L} = \sum \vec{r} \times m\vec{v}, \qquad \vec{v} = \vec{\omega} \times r$$

↳ $L = r\,m\,v_t = r\,m\,(\omega r) = (r^2 m)\,\omega = I\,\omega$

Cross Product) $\vec{a} \times \vec{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x)$

Solve) Method 1 : Directly from the definition

Method 2 : With the Inertia Matrix

Method 1 : Directly solve from the definition)

i) $\vec{v}_i = \vec{\omega} \times \vec{r}_i$ / $\vec{L}_i = \vec{r}_i \times m_i \vec{v}_i$

ii) $\vec{a} \times \vec{b} = (a_y b_z - a_z b_y, a_z b_x - a_x b_z, a_x b_y - a_y b_x)$

iii) $\vec{r}_1 = (1,0,1)\,m$, $\vec{r}_2 = (-1,0,-1)\,m$, $m_1 = m_2 = 1\,kg$, $\vec{\omega} = (0,0,2)$

① Solve $\vec{v}_1$ and $\vec{v}_2$ using $\vec{v}_i = \vec{\omega} \times \vec{r}_i$

1) m_1 : $\vec{v}_1 = (0,0,2) \times (1,0,1) = (0,2,0)$

$= (0 \cdot 1 - 2 \cdot 0,\ 2 \cdot 1 - 0 \cdot 1,\ 0 \cdot 0 - 0 \cdot 1)$

2) m_2 : $\vec{v}_2 = (0,0,2) \times (-1,0,-1) = (0,-2,0)$

$= (0 \cdot -1 - 2 \cdot 0,\ 2 \cdot -1 - 0 \cdot -1,\ 0 \cdot -1 - 0 \cdot 0)$

② Solve $\vec{L}_1$ and $\vec{L}_2$ using $\vec{L}_i = \vec{r}_i \times m_i \vec{v}_i$

1) m_1 : $\vec{L}_1 = (1,0,1) \times 1(0,2,0) = (-2,0,2)$

2) m_2 : $\vec{L}_2 = (-1,0,-1) \times 1(0,-2,0) = (-2,0,2)$

③ Combine

$\vec{L} = \vec{L}_1 + \vec{L}_2 = (-2,0,2) \oplus (-2,0,2) = (-4,0,4)\ kg \cdot m^2/s$

Method 2 : Solve with Inertia Matrix)

↳ $$I = m \begin{bmatrix} y^2+z^2 & -xy & -zx \\ -xy & z^2+x^2 & -yz \\ -zx & -yz & y^2+z^2 \end{bmatrix}$$

1) Solve I_1 and I_2

↳ m_1 at $(1,0,1)$:

$$I_1 = 1 \cdot \begin{bmatrix} 0^2+1^2 & -1\cdot 0 & -1\cdot 1 \\ -1\cdot 0 & 1^2+1^2 & -0\cdot 1 \\ -1\cdot 1 & -0\cdot 1 & 1^2+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

⇒ same for I_2

2) Combine

$$I = I_1 + I_2 = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 4 & 0 \\ -2 & 0 & 2 \end{bmatrix}$$

3) Compute $\vec{L} = I\vec{\omega}$

$$\vec{L} = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 4 & 0 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 0 \\ 4 \end{bmatrix}$$

4) Interpret

$L_x = -4$: the leak, spinning about z also produce angular momentum along $-x$ because the dumbbell is tilted

$L_y = 0$: No leak along y

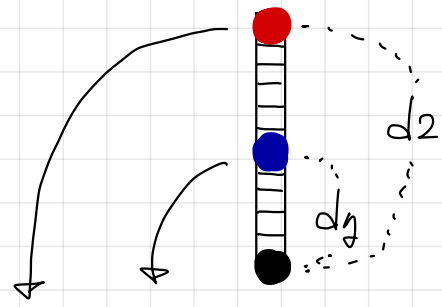
$L_z = +4$: the direct twirl, each mass circles z counterclockwise

$L_z = xv_y = 1 \cdot 2 = 2$ per mass

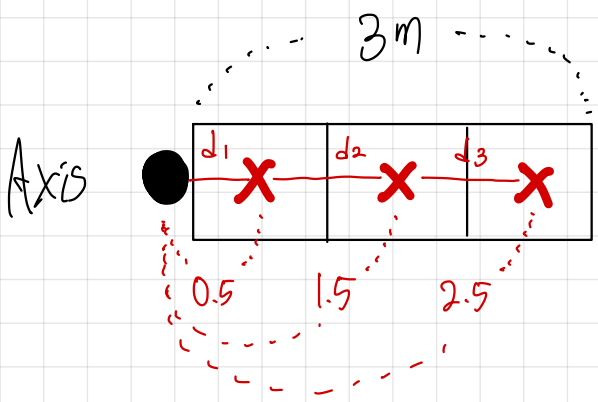
Inertia about an Axis

$$I = \sum_i m_i d^2$$

↳ distance from the axis



ex) A 3m and 3kg rod in 3 pieces



$m_1 = 1kg$ $m_2 = 1kg$ $m_3 = 1kg$

$$\Rightarrow I = (1kg \cdot (0.5m)^2) + (1kg \cdot (1.5m)^2) + (1kg \cdot (2.5m)^2)$$

Center of Mass

$$\vec{r}_{com} = \frac{\sum_i m_i \vec{r}_i}{M}$$

- m_i : mass of piece i [kg]
- $\vec{r}_i$: position of piece i measured from the chosen origin [m]
- $m_i \vec{r}_i$: weighted position [kg·m]

⇒ Center of mass is the mass-weighted average position

⇒ Heavier pieces pull the average toward themselves

⇒ Origin at the center of mass gives $\sum m_i \vec{r}_i = \vec{0}$

ex) 1 kg at $x = 0$ and 2kg at $x = 3$ m

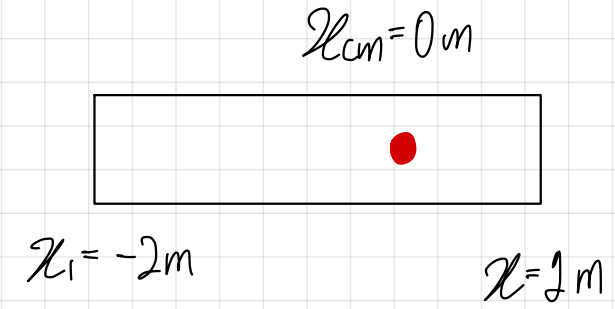
- $\sum m_i x_i = 1 \cdot 0 + 2 \cdot 3 = 6$ kg·m

$$M = 1 + 2 = 3 \text{ kg}$$

$$r_{cm} = \frac{6}{3} \text{ m} = 2 \text{ m}$$

- origin at the Center of Mass

$$\sum m_i x_i = 1 \cdot -2 + 2 \cdot 1 = 0 \text{ kg·m}$$



Parallel-Axis Theorem

If I_{com} is the moment of inertia about an axis through the center of mass

then about any parallel axis a distance d away:

$$I = I_{com} + Md^2 \qquad [kg \cdot m^2]$$

↳ M : the total mass [kg], $M = \sum_i m_i$

↳ d : the distance betw the two parallel axes [m]

↳ Swing a hammer by the middle of the handle is easy, by the end is harder. Moving the axis away from the balance point costs an extra Md^2 as if the whole mass sat at distance d

0) setup

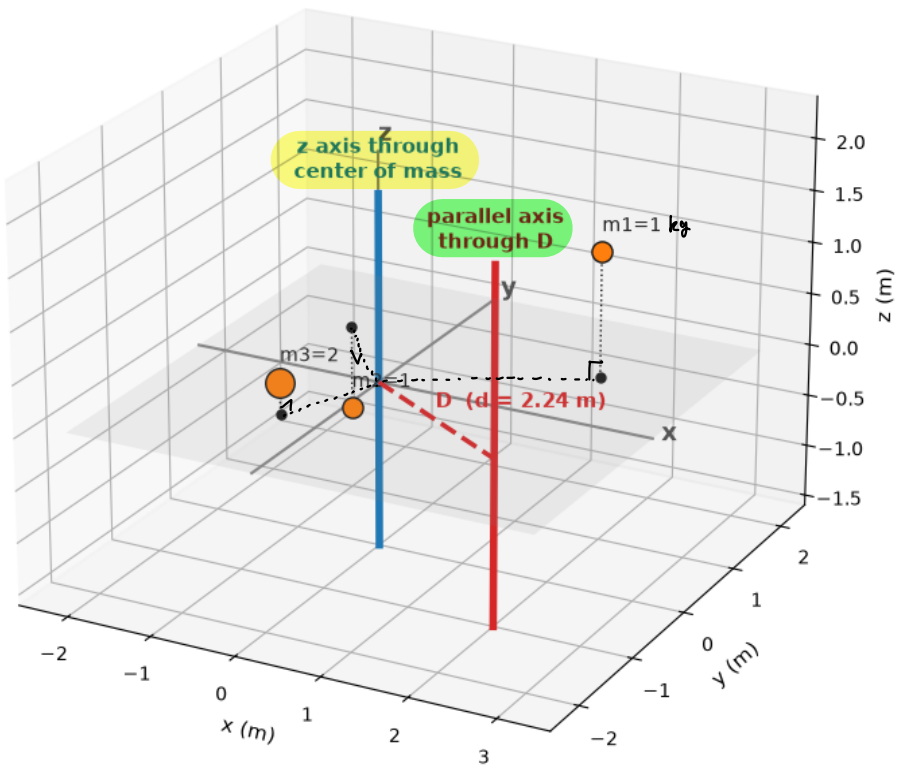
i) Axis through the COM: the z-axis, set the COM to be the origin (0,0,0)

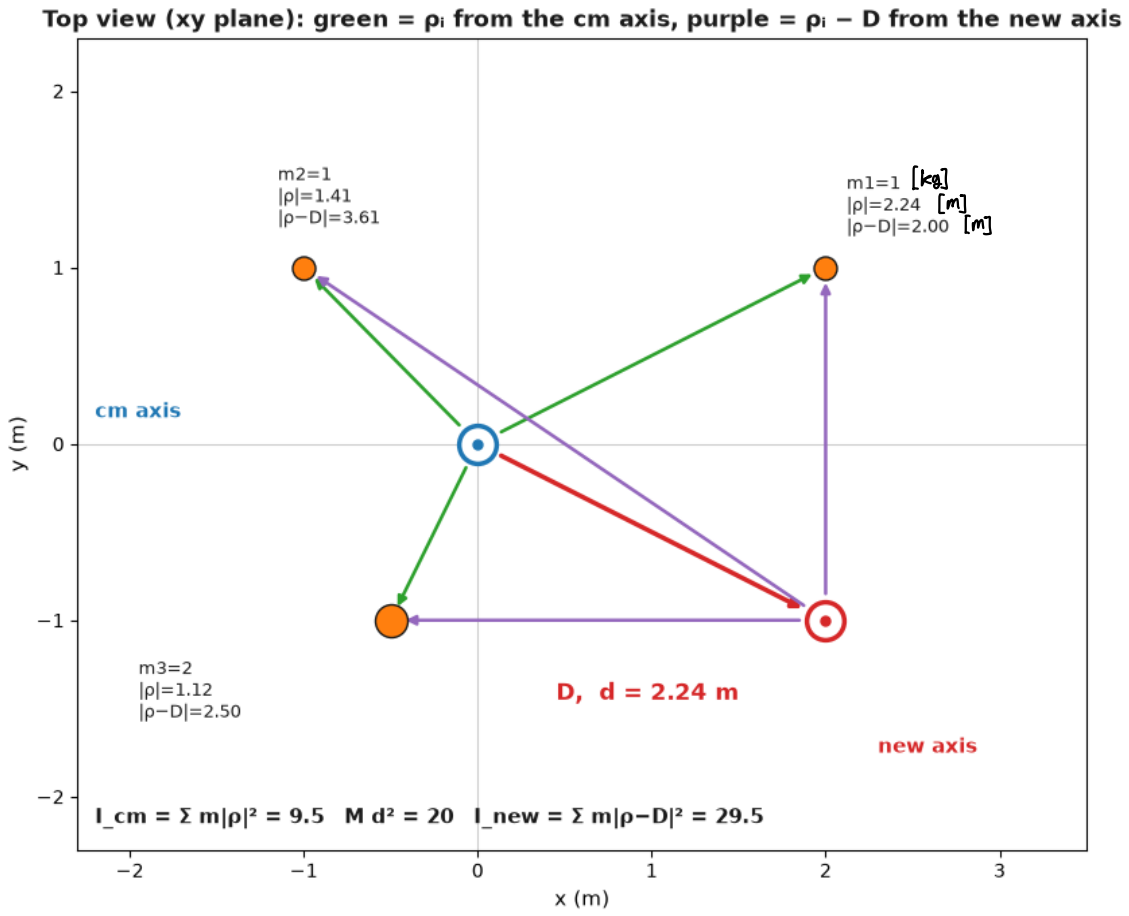
ii) To find how far a mass is from a vertical z axis, we only need the xy plane

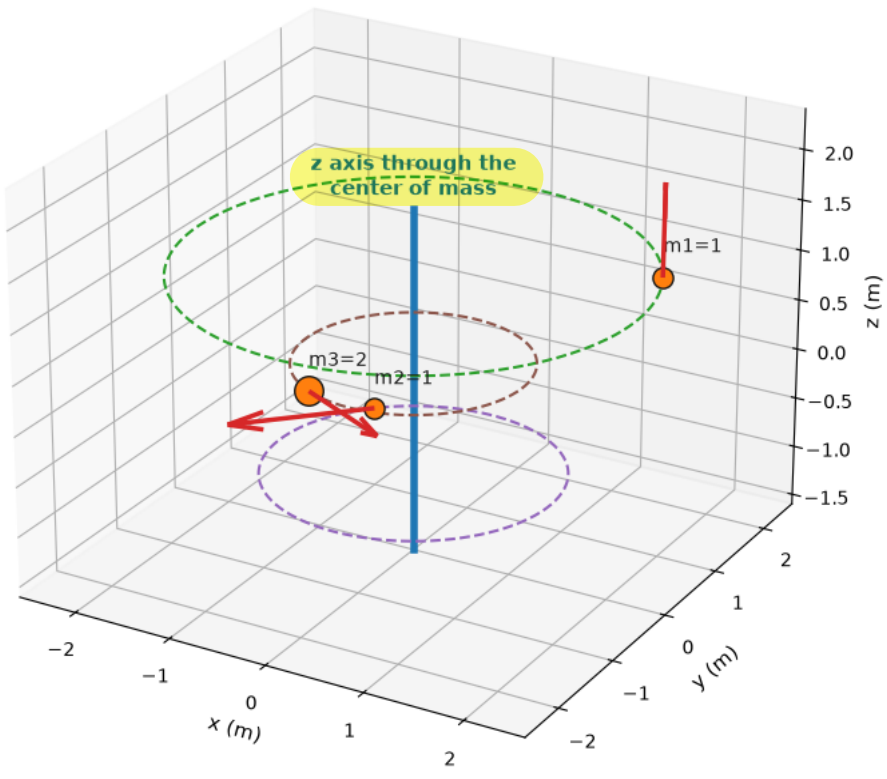
$\vec{\rho}_i = (x_i, y_i)$ so $I_{cm} = \sum_i m_i |\rho_i|^2$

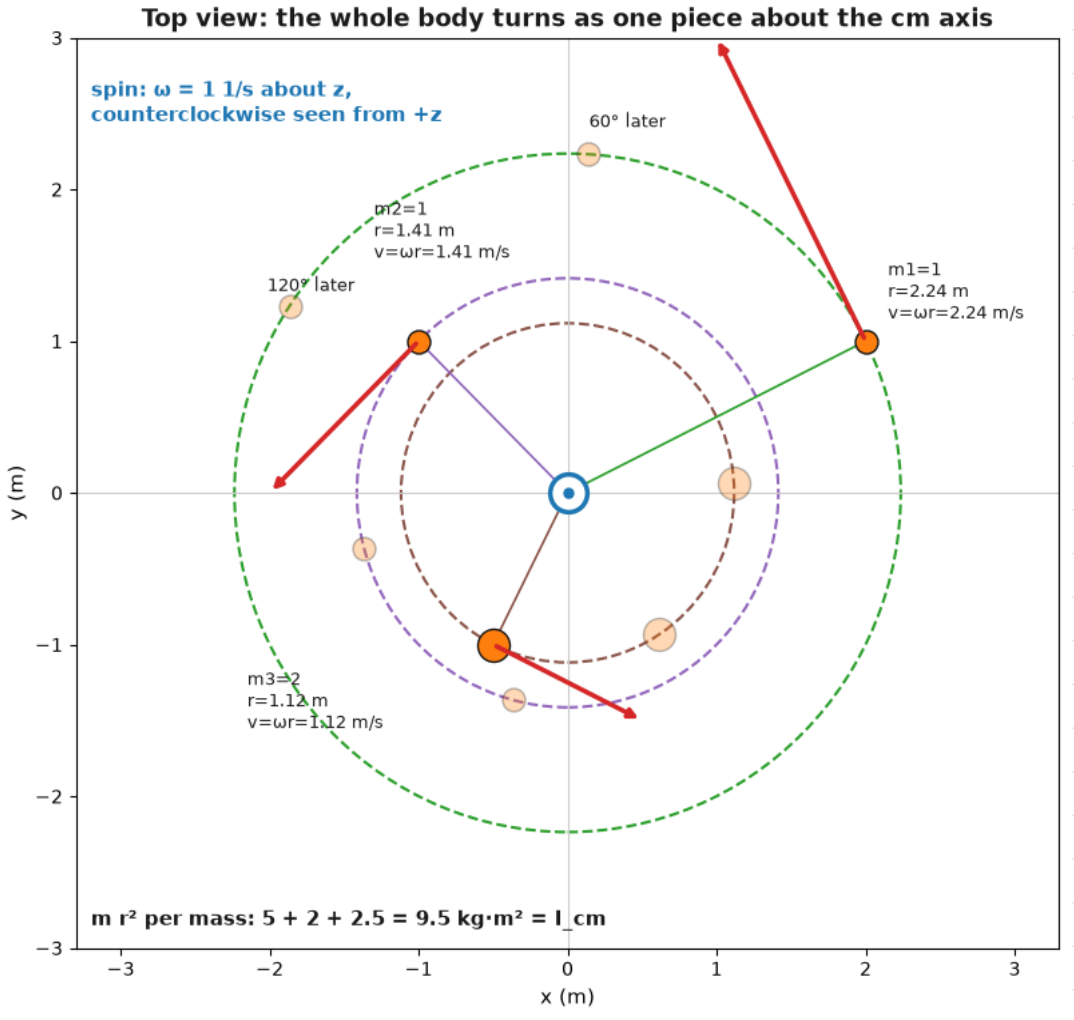
iii) Second axis: parallel to z, through $\vec{D} = (D_x, D_y)$ in the xy plane

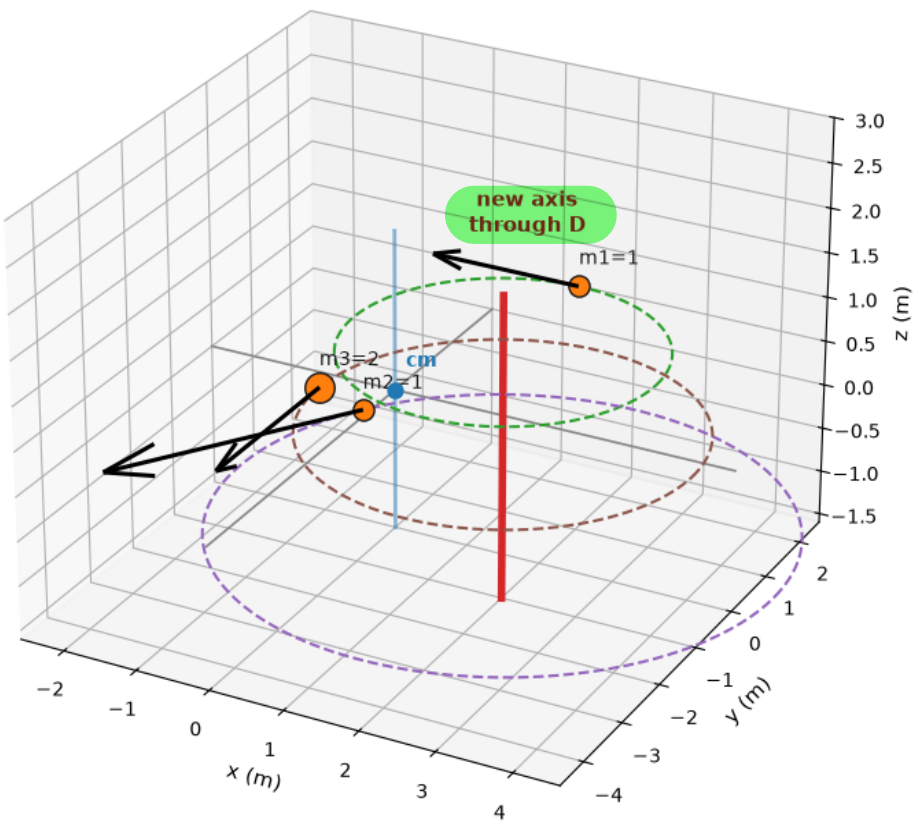
with $|\vec{D}| = d$

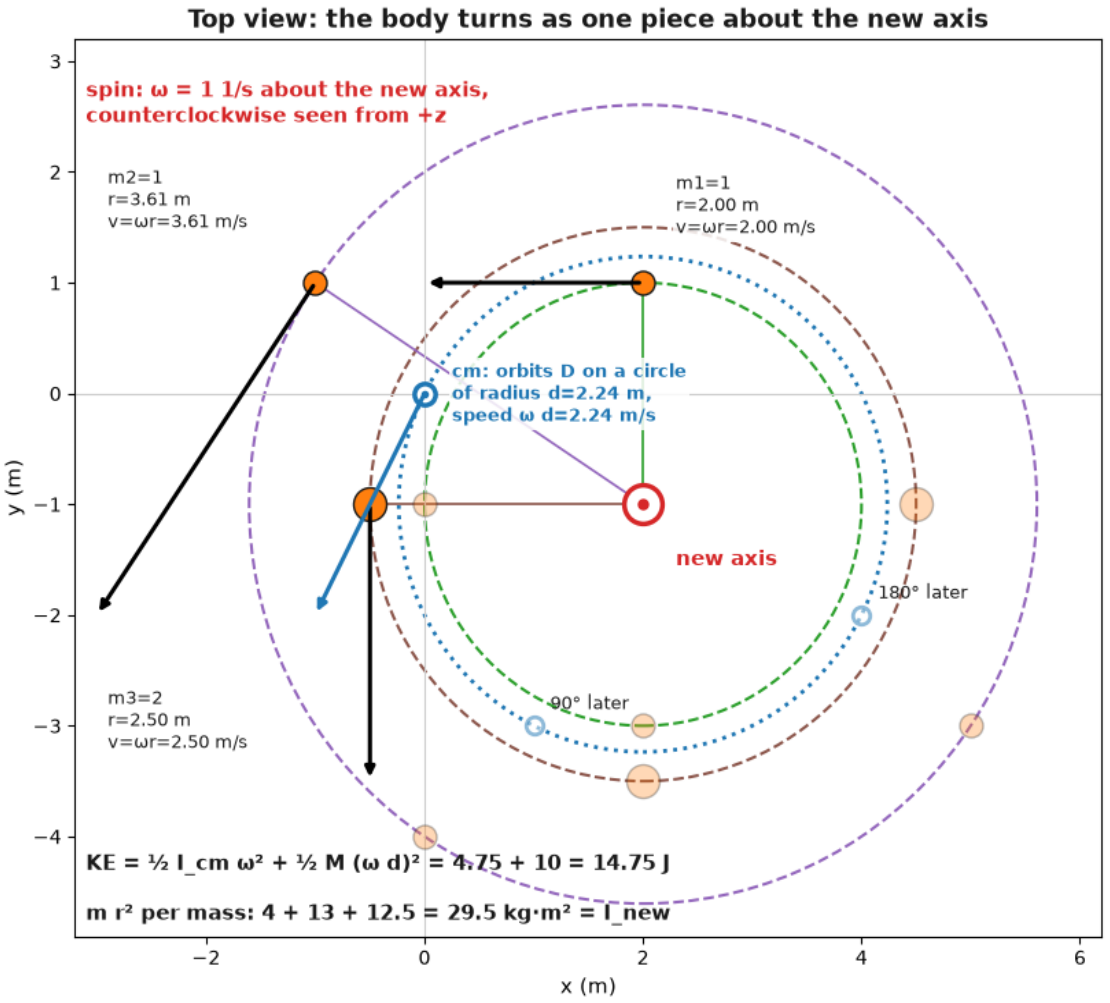

3D: two parallel axes, masses at different heights z
(dotted drops show the xy position of each mass)
z axis through
center of mass
parallel axis
through D
m1=1 kg
m3=2
m2=1
D (d = 2.24 m)
z
y
x
x (m)
y (m)
z (m)


Top view (xy plane): green = ρᵢ from the cm axis, purple = ρᵢ − D from the new axis
m2=1
|ρ|=1.41
|ρ−D|=3.61
m1=1 [kg]
|ρ|=2.24 [m]
|ρ−D|=2.00 [m]
cm axis
m3=2
|ρ|=1.12
|ρ−D|=2.50
D, d = 2.24 m
new axis
I_cm = Σ m|ρ|² = 9.5 M d² = 20 I_new = Σ m|ρ−D|² = 29.5
x (m)
y (m)


3D: every mass runs on a flat circle around the z axis
(red arrow = its velocity, tangent to the circle)
z axis through the
center of mass
m1=1
m3=2
m2=1
x (m)
y (m)
z (m)


Top view: the whole body turns as one piece about the cm axis
spin: ω = 1 1/s about z,
counterclockwise seen from +z
60° later
m2=1
r=1.41 m
v=ωr=1.41 m/s
120° later
m1=1
r=2.24 m
v=ωr=2.24 m/s
m3=2
r=1.12 m
v=ωr=1.12 m/s
m r² per mass: 5 + 2 + 2.5 = 9.5 kg·m² = I_cm
x (m)
y (m)


3D: same masses, circles now centered on the new axis
(black arrow = velocity, tangent to its circle)
new axis
through D
m1=1
m3=2
cm
m2=1
x (m)
y (m)
z (m)


Top view: the body turns as one piece about the new axis
spin: ω = 1 1/s about the new axis,
counterclockwise seen from +z
m2=1
r=3.61 m
v=ωr=3.61 m/s
m1=1
r=2.00 m
v=ωr=2.00 m/s
cm: orbits D on a circle
of radius d=2.24 m,
speed ω d=2.24 m/s
new axis
180° later
90° later
m3=2
r=2.50 m
v=ωr=2.50 m/s
KE = ½ I_cm ω² + ½ M (ω d)² = 4.75 + 10 = 14.75 J
m r² per mass: 4 + 13 + 12.5 = 29.5 kg·m² = I_new
x (m)
y (m)

1) Distance from the new axis

The vector from the new axis to mass i is $\vec{\rho}_i - \vec{D}$ so

$$I = \sum_i m_i |\vec{\rho}_i - \vec{D}|^2$$

2) Expand the squared distance

$$|\vec{\rho}_i - \vec{D}|^2 = |\vec{\rho}_i|^2 - 2\vec{\rho}_i \cdot \vec{D} + |\vec{D}|^2$$

3) Split the Sum, $|\vec{D}| = d$

$$I = \underbrace{\left(\sum_i m_i |\vec{\rho}_i|^2\right)}_{I_{cm}} - 2\underbrace{\left(\sum_i m_i \vec{\rho}_i\right)}_{0} \vec{D} + \underbrace{\left(\sum_i m_i\right)}_{M} \cdot d^2$$

$$\therefore \quad I = I_{cm} + Md^2$$

↳ The moment of inertia about an axis equals the moment of inertia about the parallel axis through the COM plus Md^2

- When do we use this?

Input: I_{com}, M, d

$\Downarrow$

Output: I_{pivot}

$$I_{pivot} = I_{com} + Md^2$$

$$= M\frac{L^2}{12} + M\left(\frac{L}{2}\right)^2$$

$$= M\frac{L^2}{3}$$

(We have) Rod mass, Rod length, Moment of inertia about axis through the COM perpendicular to the rod, and where the rod actually turns (the pivot at one end, $d = L/2$)

Question) If the pivot is on one end and we know where what is the moment of inertia for the pivot?

$\omega_{unstable}$

(Context) We balance the rod upright on its pivot, then tilt it by a tiny angle. gravity pulls it further over, and the further over it is, the harder gravity pulls. So the tilt grows on its own

(Question) How fast the tiny tilt grows?

(We want) One number $\omega_{unstable}$ [1/s] that says the tilt grows like

$$e^{\omega_{unstable} t}$$

- $\Omega_{unstable}$ is the growth rate of a small tilt of the upright rod

ansatz: $\theta(t) \propto e^{\omega_{unstable} t}$

↳ θ: the rod tilted by θ (rad) from vertical

$$\omega_{unstable} = \sqrt{\frac{3}{2}\frac{g}{L}}$$

↳ L: Rod Length

We have) Mass $m = 1\,[kg]$, length $L = 1\,[m]$, gravity $g = 9.81\left[\frac{m}{s^2}\right]$

We know) Moment of Inertia about the pivot $I_{pivot} = \frac{mL^2}{3}$ $[kg \cdot m^2]$

Torque $\tau = r F \sin(\theta) = I \alpha$ $[N \cdot m]$

Location of the Center of Mass $\vec{r}_{com} = \frac{\sum_i m_i \vec{r}_i}{M}$

We are using $I_{pivot} = \frac{1}{3} mL^2$

We want) $\omega_{unstable}$ in $\theta \propto e^{\omega_{unstable} t}$ $[1/s]$

$\Rightarrow$ 1) find the torque gravity puts on the tilted rod,

2) turn the torque into an angular acceleration,

3) then solve for how the tilt evolves

1) the torque of gravity

↳ We have the rod, tilted by θ (rad) from vertical. We want to find the torque about the pivot because torque is the only thing that changes how the rod turns

i) Gravity acts as one force. $F = ma = mg$ [N]

pulling straight down at the COM

ii) COM is $\vec{r}_{com} = \frac{\sum_i m_i \vec{r}_i}{M}$ and in our case $\vec{r}_{com} = \frac{L}{2}$

Where the rod is uniform and no extra mass

$\Rightarrow dm = \frac{m}{L} dr$: $r_{com} = \frac{1}{m} \int_0^L r \frac{m}{L} dr = \frac{1}{L} \cdot \frac{L^2}{2} = \frac{L}{2}$

iii) The angle between the rod and straight down is $180° - \theta$

or $\pi - \theta$ and its sine equals $\sin(\pi - \theta) = \sin(\theta)$

$\Rightarrow \tau = r F \sin(\varphi) = \frac{L}{2} \cdot mg \cdot \sin(\theta) = \frac{mgL}{2} \sin\theta$

② The equation of Motion

↳ $\tau = \frac{mgL}{2}\sin\theta$, $I_{pivot} = \frac{1}{3}mL^2$, $\tau = I\alpha$

↳ we know τ and I_{pivot} from $\tau = I\alpha$ and $\alpha = \ddot{\theta}$ [g/s^2]

is what we want

$$\alpha = \frac{\tau}{I_{pivot}} = \frac{\frac{mgL}{2}\sin\theta}{\frac{1}{3}mL^2} = \frac{3}{2}\frac{g}{L}\sin\theta = \ddot{\theta}$$

- For a small θ, $\sin\theta \approx \theta$

eg) at $5^\circ = 0.0873$ rad, $\sin\theta = 0.0872$, an error rate of about 0.1%

↳ Currently $\alpha = \ddot{\theta} = \frac{3}{2}\frac{g}{L}\sin\theta$ and we want a linear equation

we can solve directly

$$\therefore \alpha = \ddot{\theta} = \frac{3}{2}\frac{g}{L}\sin\theta \approx \frac{3}{2}\frac{g}{L}\theta$$

(3) Solve how the tilt evolves

↳ we have $\ddot{\theta} \approx \frac{3}{2}\frac{g}{L}\theta$ and we want the growth rate

to try $\theta = e^{\omega t}$ and see which ω works

↳ we know $\ddot{\theta} = \omega^2 e^{\omega t}$ so $\omega^2 = \frac{3}{2}\frac{g}{L}$

ans $\omega_{unstable} = \sqrt{\frac{3}{2}\frac{g}{L}}$

$$\frac{dx}{dt} = \alpha x \quad \text{and} \quad x(t) = x_0 e^{\alpha t}$$

Most physics cares "How does x changes" Not "what is x"

↳ The simplest law of change is $\frac{dx}{dt} = \alpha x$

$\Rightarrow$ The rate of change of x is proportional to its current size

ex) More tilt gives more gravity torque, which gives more angular acceleration

↳ The exponential is forced by the law rate & size

1) Separation of variables

$$\frac{dx}{dt} = \alpha x \rightarrow \frac{1}{x} dx = \alpha dt$$

2) Integrate

$$\int \frac{1}{x} dx = \int \alpha \, dt \rightarrow \ln x + C_1 = \alpha t + C_2$$

$$\rightarrow \ln x = \alpha t + C$$

3) exponentiate

$$\ln x = \alpha t + C$$

$$\longrightarrow \quad e^{\ln x} = e^{\alpha t + C_{12}} = e^{\alpha t} \cdot \underbrace{e^{C}}_{x_0 \equiv \text{Starting Value}}$$

$$\longrightarrow \quad x(t) = x_0 \, e^{\alpha t}$$

- Same idea for the rod, but in higher order

↳ The rod obeys **acceleration ∝ size** not rate ∝ size

$$\overset{\circ\circ}{\theta} = k\theta$$

i) $\dot{x} = a x$: first order → rate ∝ size

ii) $\ddot{x} = a x$: second order → acceleration ∝ size

↳ **Why exponential ?**

exponential reproduces itself when differentiated

$$\frac{d}{dt} e^{\omega t} = \omega e^{\omega t} \quad \rightarrow \quad \frac{d}{dt} \omega e^{\omega t} = \omega^2 e^{\omega t}$$

↳ Why linear ?

We made $\alpha = \ddot{\theta} = \frac{3}{2}\frac{g}{L}\sin\theta$ (non-linear) $\Rightarrow$ $\frac{3}{2}\frac{g}{L}\theta$ (linear)

- Linearity : the unknown θ (and its derivative) only appears to the first power, multiplied by constants

$\Rightarrow$ Linear equations have superposition

If θ_1 and θ_2 are solutions, then $a\theta_1 + b\theta_2$ is a solution too

↳ we could solve for simple pieces $e^{+\omega t}$ and $e^{-\omega t}$, then scale and add them to match any start

↳ the growth rate ω does not depend on the size of the tilt

- Linearizing : replace the curve by its tangent line at the equilibrium

ex) $f(\theta) = k\sin\theta$ at the starting point $\theta = 0$ (upright position)

$f(\theta) = f(0) + f'(0)\,\theta$

$f(0) = k\sin(0)$: equilibrium

$f'(0) = k\cos(0)$

$= 0 + k\cos(0)\,\theta$

$= k\theta$

- We have a law that the rod's tilt must obey and a starting situation

1) Law: $\ddot{\theta} = k\theta$ where $\alpha = \ddot{\theta} = \frac{3}{2}\frac{g}{L}\theta$

2) Start: $\theta(0) = \theta_0$ [rad] and $\dot{\theta}(0) = 0$ [rad/s]

3) Goal: find the function $\theta(t)$ that obeys the law and matches the start

↳ find $\theta(t)$)

i) Set ansatz $\theta(t) = C e^{\omega t}$ (the solution we are looking for)

ii) Differentiate twice: $\ddot{\theta} = C\omega^2 e^{\omega t}$ Put both into $\ddot{\theta} = k\theta$

$$C\omega^2 e^{\omega t} = k C e^{\omega t} \Rightarrow \omega^2 = k = \frac{3}{2}\frac{g}{L} = \frac{3}{2}\frac{9.81}{1} = 14.715$$

iii) $\omega = \pm\sqrt{k} \Rightarrow$ Set $\omega \equiv \sqrt{k}$ so that we can use $\pm\omega$

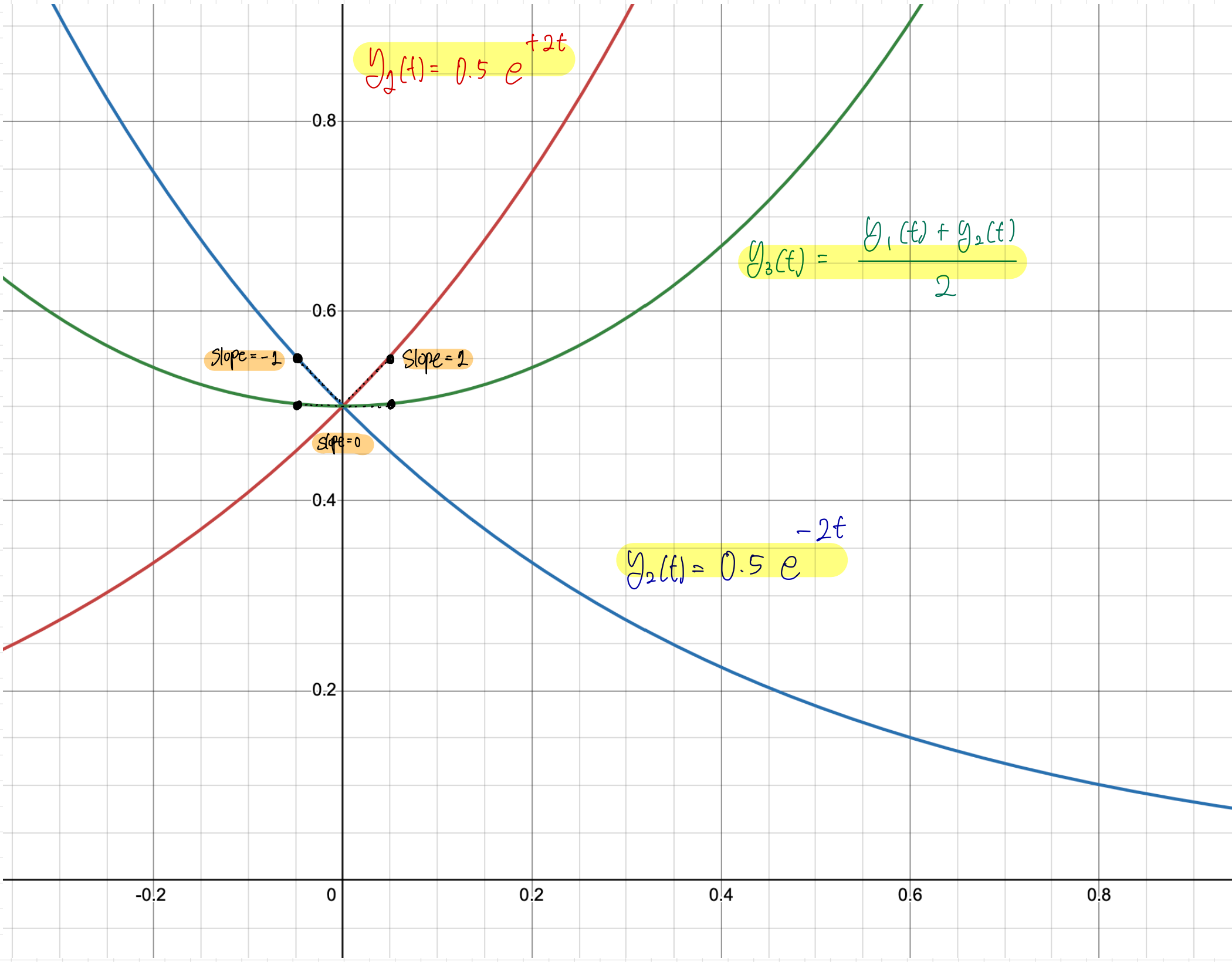
$= \pm\sqrt{14.715} = \pm 3.836 \Rightarrow$ We have $+\omega$ and $-\omega$

iv) Since second order we have two roots: $+\omega$ and $-\omega$

$\Rightarrow$ two solutions: $\theta_1(t) = \theta_{0,1} e^{+\omega t} = A e^{+\omega t}$, $\theta_2(t) = \theta_{0,2} e^{-\omega t} = B e^{-\omega t}$

a) $e^{+\omega t}$: grows $\Rightarrow (+\omega)^2 e^{+\omega t} = k e^{+\omega t}$

b) $e^{-\omega t}$: decays $\Rightarrow (-\omega)^2 e^{-\omega t} = k e^{-\omega t}$



- Starting points $(t=0)$

i) $y_1(t) = \underset{A}{0.5}\ e^{+2t}$

↳ Height at $t=0$: $\underset{A}{0.5}$

↳ Slope at $t=0$: $y_1'(t) = 0.5 \times 2\ e^{2t}$

$= 1\ e^{2t}$

$= 1 \cdot y_1(t)$

↳ Not true for our system

Since it should be 0 at $t=0$

ii) $y_2(t) = \underset{B}{0.5}\, e^{\underset{}{-2t}}$

↳ Height at $t=0$: $\underset{B}{0.5}$

↳ Slope at $t=0$: $y_1'(t) = 0.5 \times -2\, e^{2t}$

$= -1\, e^{2t}$

$= -1 \cdot y_1(t)$

↳ Not the for our system

iii) $y_3(t) = \dfrac{y_1(t) + y_2(t)}{2}$ Since it should be 0

↳ Height at $t=0$: $\underset{A}{0.5} + \underset{B}{0.5} = 0.5$

↳ Slope at $t=0$: $w(\underset{A}{0.5} - \underset{B}{0.5}) = 0$

The for our system

- $\theta(t) = A e^{+wt} + B e^{-wt}$

↳ We need a history that matches the start $(\theta_0, \dot{\theta}_0)$

tilt , tilt rate

ex) the rod is releases from rest at 5° ⇒ $(5°, 0)$

⇒ so both $\theta_0 = 5°$ and $\dot{\theta}_0 = 0$ should be met

⇒ the start has two requirements

⇒ two requirements need two knobs

↳ one block $Ce^{\omega t}$ only give one knob $\left(\frac{d}{dt}e^{\omega t} = \omega e^{\omega t}\right)$

at $t=0$, it determines both numbers $\text{tilt} = C$ and $\text{rate} = \omega C$

But the rate is locked to the tilt C since rate is just scaled C

ex) We need $(C = 5°,\ \omega C = 0)$ where $\omega = 3.84$

$C=5 \rightarrow C=5\ (\checkmark)$ $\omega C = 19.2 \neq 0\ (X)$

∴ fixing one number breaks the other

↳ For second-order law with two blocks $e^{\lambda_1 t}$ and $e^{\lambda_2 t}$

$\theta(t) = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$

$\theta_0 = A + B$ $\dot{\theta}_0 = \lambda_1 A + \lambda_2 B$

$\Rightarrow$ tilt at $t=0$ $(e^0 = 1)$: $\theta(0) = A(1) + B(1) = A+B$

$\Rightarrow$ rate at $t=0$ $\left(\frac{d}{dt}e^{\lambda t} = \lambda e^{\lambda t}\right)$: $\dot{\theta}(0) = \lambda_2 A + \lambda_2 B$

$\Rightarrow$ for our case $\theta(t) = Ae^{+\omega t} + Be^{-\omega t}$

$\theta_0 = A + B$ $\dot{\theta}_0 = \omega(A-B)$

↳ $A = \frac{\dot{\theta}_0 - \lambda_2 \theta_0}{\lambda_1 - \lambda_2}$ $B = \frac{\lambda_1 \theta_0 - \dot{\theta}_0}{\lambda_1 - \lambda_2}$

- For a second-order law that is linear, homogeneous, and has two independent solution θ_1 and θ_2, the general solution is

$$\theta(t) = A\,\theta_1(t) + B\,\theta_2(t)$$

↳ Linear so superposition, so the sum is solution

↳ Since two independent blocks and linear, the weighted sum is the general solution

↳ $\theta_0 = A + B$, $\dot{\theta}_0 = \lambda_1 A + \lambda_2 B$